Sound level meter
TYPE 6236

Instruction manual

ACO Co.,Ltd.

Components of this Instruction manual

This instruction manual refers to the functions of, and operating instructions for, Sound Level Meter TYPE 6236 (abbreviated to "this equipment" in what follows)

This instruction manual consists of following chapters.

Outline
The components, characteristics, block diagram of this equipment are described

Locations and their functions
The names and functions of keys and terminals are briefly described.

Liquid crystal screen
The symbols displayed on the screen are described.

Preparation
The power supply, check before use, installation of this equipment, connection of cables and various key setting are described.

Measurement
Basic idea of measurement method is described.

Recording
How to save or recall data is described.

Printing/Collecting
How to print or collect the measurement data is described.

Output terminal
Output terminal of this equipment is described.

Specification
The specification of this equipment is described.

Safety precautions

To prevent bodily injury or damage to property, the following safety precautions must be observed.

This manual contains important safety and operating instructions for this equipment.

Read all instructions, before using the instrument.

After reading all instructions, keep this manual for quick reference

1. Expressions of safety instructions

⚠ WARNING
Calls attention to a procedure, practice, or condition that could possibly cause death or bodily injury.

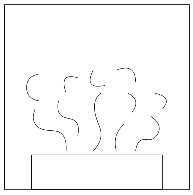
⚠ CAUTION
Calls attention to a procedure, practice, or condition that could possibly cause bodily injury or damage to instrument.

NOTE
It is an advisory explanation to use this equipment correctly. (It is not a safety instruction)

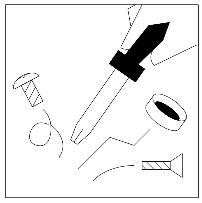
2. Important safety instructions

⚠ WARNING

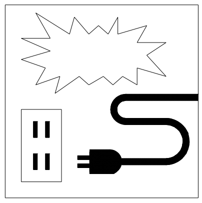
Stop using the instrument, when producing smoke, bad smell or noise.
It causes fire or shock hazard.
Turn off the POWER switch and unplug the AC adaptor (optional) from outlet as soon as possible.
To reduce risk of injury, take it to a qualified serviceman when service or repair is required.
Please contact ACO co. or the dealer when service or repair is required.



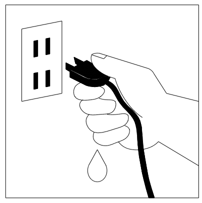
Do not substitute parts or modify instrument.
It causes bodily injury, fire or shock hazard.



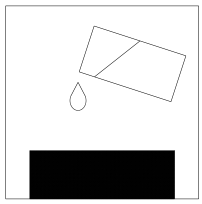
Do not use the AC power adaptor except the optional AC-1026.
Other type of adaptor may cause damage to the instrument.



Do not touch the plug of AC adaptorwith wet hands.
It causes shock hazard.

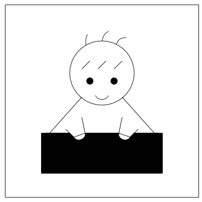
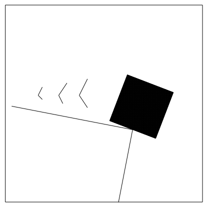
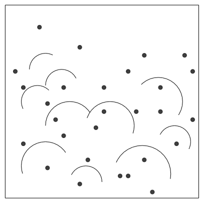
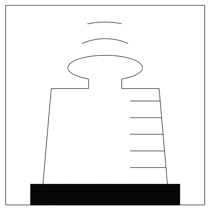
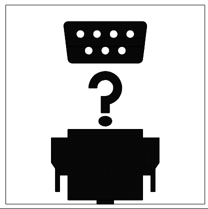
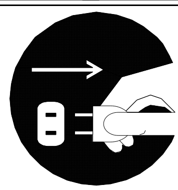
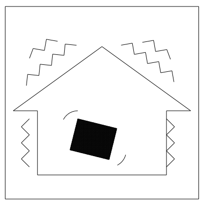


Stop using the instrument, when an object or liquid falls/spills into the instrument.
It causes fire or shock hazard.
Turn off the POWER switch and unplug AC adaptor (optional) from outlet as soon as possible.
To reduce risk of injury, take it to a qualified serviceman when service or repair is required.
Please contact ACO co. or the dealer when service or repair is required.



3. Cautions for usage

This equipment is assembled with precision parts.
To prevent bodily injury or damage to the instrument, the following cautions must be observed.

⚠ CAUTION	
Keep the instrument away from the children. If the instrument falls down, it is very dangerous.	
Do not place it on an unstable place (shaky table or sloping place). If the instrument falls down, it is very dangerous.	
Do not expose the instrument to moisture or dust. It causes fire or shock hazard.	
Do not put heavy objects on the instrument. It causes damage to the instrument.	
Connect cable properly, it is instructed in this manual. Wrong connection causes fire hazard.	
Before you move the instrument to other place, turn off the POWER switch and remove all wiring.	
Do not put the instrument on the vibrating place. If the instrument falls down, it is very dangerous.	
For avoiding liquid spill, remove alkaline dry batteries when you don’t use for long period of time. It is recommended to remove alkaline dry batteries after each use.	

Disclaimer in usage of the software product

When this software is used, it is assumed that the customer has accepted all the following items.

1) The customer is permitted to use this software product based on the agreement of use conditions, not to transfer or sell to the third party.
   In case the customer cannot accept the following items, the product cannot be cleared to use, either.

2) The software product, together with attached documents such as instruction manuals, belongs to Aco Ltd. and is protected by the Copyright Law., etc.
   The customer is not permitted either to copy, modify, alter this software product, or remove the product label.
   The customer is not permitted to create any similar products, or have the third party do these actions.

3) Please do try hard to keep every user or users scheduled about the items above before the use of this product.
   As would be realized, the customer may be considered to have acted against the agreement when the user of this product acted against it.

The Quantifier form of International standard and JIS (Japanese Industrial Standards).

The Quantifier is excerpted from ISO 1996、3891、IEC 60804、JIS Z 8202、8731.

<table>
<tr><th colspan="2">Notation of TYPE6236</th><th>Name</th><th>Frequency weighting characteristics</th><th colspan="2">ISO</th><th>IEC</th><th colspan="2">JIS</th></tr>
<tr><td colspan="2">L_A</td><td>A-weighted sound pressure level</td><td>A-weighted</td><td colspan="2">L_{pA}</td><td>−</td><td colspan="2">L_{pA}</td></tr>
<tr><td colspan="2">L_C</td><td>C-weighted sound pressure level</td><td>C-weighted</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_P</td><td>Sound pressure level</td><td>Z-weighted</td><td colspan="2">L_P</td><td>−</td><td colspan="2">L_P</td></tr>
<tr><td colspan="2">L_{Aeq}</td><td>Equivalent continuous A-weighted sound pressure level</td><td>A-weighted</td><td colspan="2">$L_{Aeq,T}$</td><td>$L_{Aeq,T}$</td><td colspan="2">$L_{Aeq,T}$</td></tr>
<tr><td colspan="2">L_{Ceq}</td><td>Equivalent continuous sound pressure level</td><td>C-weighted</td><td colspan="2">−</td><td>$L_{Ceq,T}$</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_{eq}</td><td>Equivalent continuous sound pressure level</td><td>Flat</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_{AE}</td><td rowspan="3">Sound exposure level</td><td>A-weighted</td><td colspan="2">L_{AE}</td><td>L_{AE}</td><td colspan="2">L_{AE}</td></tr>
<tr><td colspan="2">L_{CE}</td><td>C-weighted</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_E</td><td>Flat</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td rowspan="5">L_{AN}</td><td>L_{A05}</td><td>5% of the percentile sound pressure level</td><td rowspan="5">A-weighted</td><td rowspan="5">$L_{AN,T}$</td><td>$L_{A5,T}$</td><td>−</td><td rowspan="5">$L_{AN,T}$</td><td>$L_{A5,T}$</td></tr>
<tr><td>L_{A10}</td><td>10% of the percentile sound pressure level</td><td>$L_{A10,T}$</td><td>−</td><td>$L_{A10,T}$</td></tr>
<tr><td>L_{A50}</td><td>50% of the percentile sound pressure level</td><td>$L_{A50,T}$</td><td>−</td><td>$L_{A50,T}$</td></tr>
<tr><td>L_{A90}</td><td>90% of the percentile sound pressure level</td><td>$L_{A90,T}$</td><td>−</td><td>$L_{A90,T}$</td></tr>
<tr><td>L_{A95}</td><td>95% of the percentile sound pressure level</td><td>$L_{A95,T}$</td><td>−</td><td>$L_{A95,T}$</td></tr>
<tr><td colspan="2">L_{Amax}</td><td>Maximum sound pressure level</td><td>A-weighted</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_{Amin}</td><td>Minimum sound pressure level</td><td>A-weighted</td><td colspan="2">−</td><td>−</td><td colspan="2">−</td></tr>
<tr><td colspan="2">L_{Cpeak}</td><td>Peak sound pressure level</td><td>C-weighted</td><td colspan="2">−</td><td>L_{Cpeak}</td><td colspan="2">−</td></tr>
</table>

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Overview

This specification refers to the Sound Level Meter TYPE6236.It covers most measurands corresponding to JIS and –ISO.The 6236, provided with many functions usually mounted in equivalent products, has been realized at an extremely low price.
Measurement of most measurands, such as Equivalent continuous A-weighted sound pressure level (L_{Aeq}), Sound exposure level (L_{AE}), A-weighted sound pressure level (L_A), etc., is possible.
The 6236 was developed to keep comfortable sound environment as well as safe and healthy life of people, both to be realized by the evaluation of environmental noise such as traffic noise or industrial equipment noise, or by better understanding of the labor health environment at offices, factories, etc.
The impressive design of 6236 symbolizes satisfactory operations and many performances related to JIS and/or IEC.　It sure is a highly efficient and highly reliable precision sound level meter, to be supported by the next generation.

Features

- Excellent cost/performance ; covers most measurands in current criteria
- The “0-dB” function 【 world first 】(option)
  ; measurement of the noise of ultra-low sound pressure level is possible.
  The function displays the greatest force in the evaluation of “quietness” or sound quality of silent model of recent IT/OA equipments, as well as in those of air-conditioning noise level or sound isolation capability of newly built concert halls where the room ambient noise is below "NC-20 .
- Percentile sound pressure level (L_N) ; any 5 selectable values is available
- Measurement of　Equivalent continuous A-weighted sound pressure level (L_{Aeq})
  ; Measurement of environmental noise required to secure occupational heal
- Wide linearity range of 100dB ; Covers wide range of 20~130Db
- Equipped with an USB Ver1.1 function ; allows data processing for PC
- Equipped with a memory function ; realized by in built-in memory or memory card.
- Backlight LCD screen for high ; visibility and easy-on-the eye display
- Timer function ; measurement can be paused or restarted at any point of time by installing the function.
- Abundant program cards(Option) ; 1/1and 1/3-octave Real-time analysis card, FTT analysis Card, RSR card (Real sound recording Card), etc.

Configuration

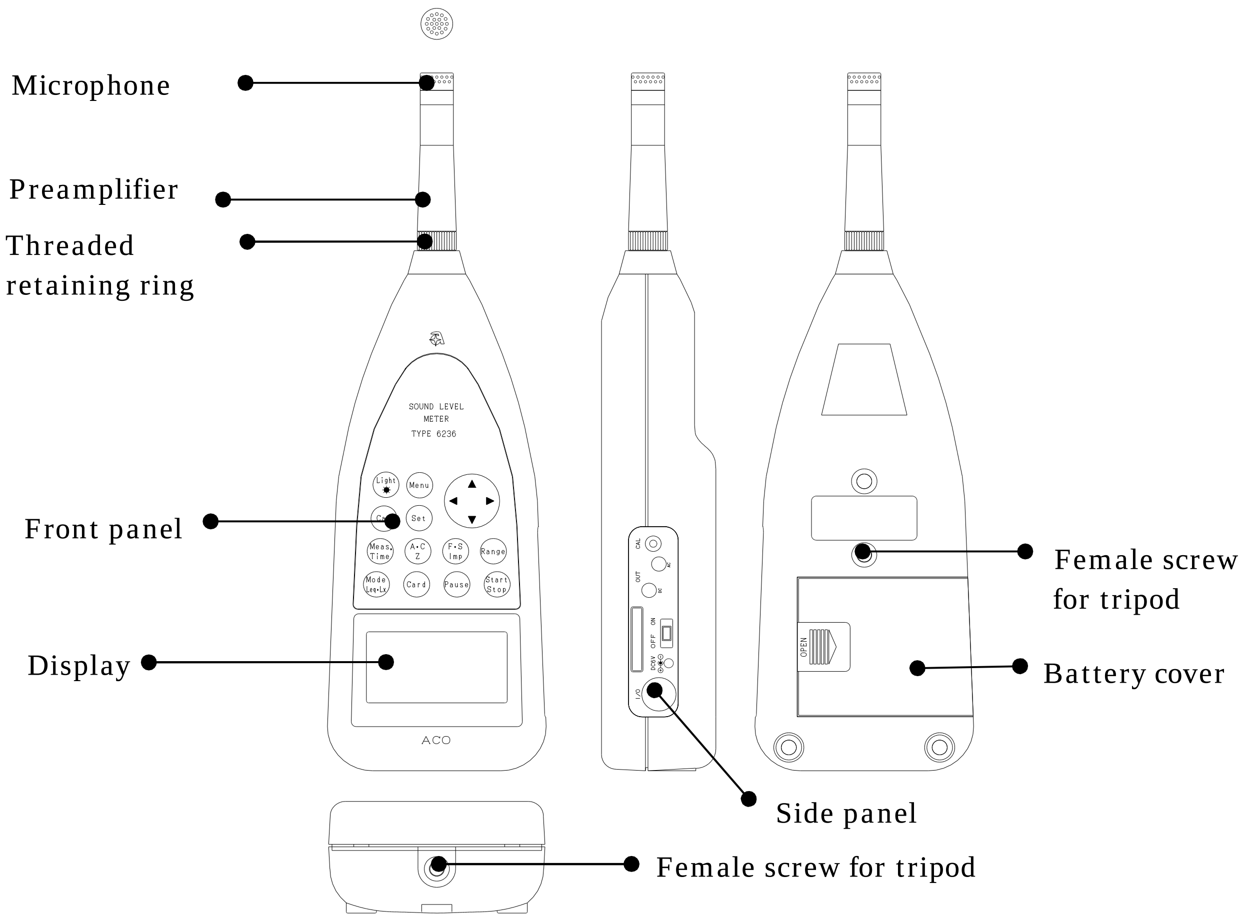
1) Sound Level Meter	TYPE6236	1
2) Memory Card (SD Card)		1
3) Windscreen(ϕ50)		1
4) Batteries	LR6	4
5) Screwdriver		1
6) Hand strap		1
7) Instruction manual		1
8) Carrying case		1

9) Option

・1/1 and 1/3-octave Real-time Analysis Card	NA-0038
・FFT Analysis Card	NA-0038F
・RSR Card (Real Sound Recording Card)	NA-0038R
・Data management software	NA-0038M
・0-dB function (0~80dB(A))	6236(0dB)
・AC adapter	AC-1026
・Output cable(BNC pin cord)	BC-0071
・USB interface cable	BC-0036
・Extension cable(2m～30m)	BC-0046-2～30
・Tripod exclusively for sound level meter	NA-0333
・Sound calibrator	TYPE2127

Locations and their functions

Front/Back/Side view of the main body:



Front

Microphone Preamplifier

The microphone and the preamplifier are comprised as one body.

They can be placed apart from the main body and connected to it with the optional extension cable

Display

It is a liquid crystal display with backlight. The sound level is displayed here with numerical value or bar graph. The operation condition of the sound level meter, setting condition of the measurement mode, various alerts, etc. are also displayed.

Back

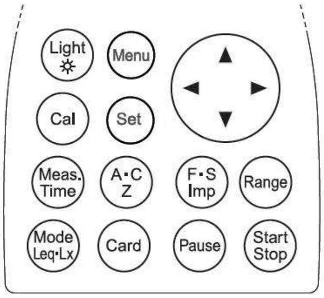
Female screw for tripod

It is possible to mount the main body to this tripod for the camera with screw.

Batteries case

Put four LR6 type Alkali dry batteries.

Function key



Light key

The backlight illuminates the display in darkness, which goes out automatically 30 seconds later or by pushing the key again.

Menu key

It is pushed to set up the measurement condition, when the display is adjusted to 1/3 page of the menu panel.
The item is selected with cursor key ▲▼, and input starts with ▶ , as well as the alteration with ▲▼. To go back to the measurement setting screen, push [SET] key again.

Cal key

When the calibration or level setting with the equipment connected, this key is used.

Set key

The key to be used to fix the input.

Meas. Time key

The key to set the measurement period (interval time terminated with a pair of Star/Stop).
It changes on pushing the key as: key is pushed again.
1s, 3s, 5s, 10s, 1m, 5m, 10m, 15m, 30m, and 1h, 8h, 12h, 24h, and *** (Forever: Until Stop).

A･C, Z key

The key to select frequency weighting A, C, and Z (FLAT)

F･S, Imp key

The key to select time-weighting Fast, Slow, and Imp

Range key

Range setting key which enables the following 6 ranges:
20～80、20～90、20～100、20～110、30～120、40～130

Mode Leq ･Lx key

The key to display the calculation results. Each push gives various calculation results selected on the Menu screen.

Card key

The key to use various option cards

Pause key

By pushing the key, the measurement is paused to eliminate any unexpected noise or anomaly during the measurement. It is resumed by pushing the key again. By using the data elimination function, it is possible to exclude the data 3 or 5 seconds before the key is pushed.

Start／Stop key

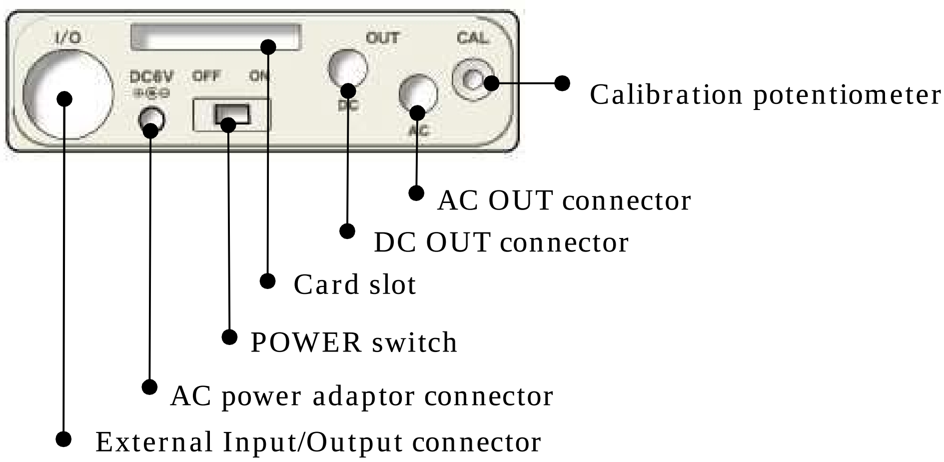
The key to start the measurement of various mode or to terminate it.

Strap

Used to prevent unexpected drop of the main body. Please put it through your wrist when you measure with the body in hand.

Side view of the main body

Side panel



AC power adaptor connector

By using the optional AC adaptor, AC100V is available for the measurement.
Please do not use any other power supplyies than specified AC adaptor. It may cause breakdown or malfunction.

AC／DC out connector

AC： outputs frequency-weighted AC signal.
DC： outputs DC level signal .

External Input/Output connector

Input or output terminal for control signal or measurement data, which can be connected to a printer, level recorder, or personal computer.

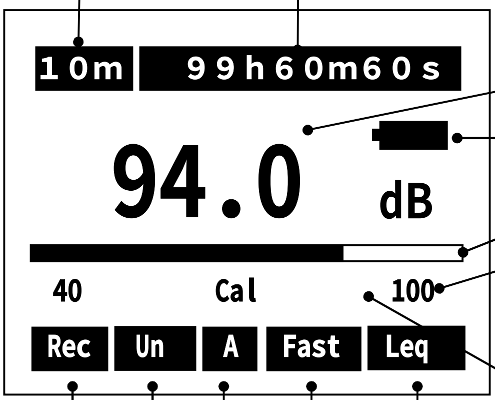
Card slot

The slot for memory card or optional program card.

NOTE
・Please watch out for the card slot portion when you have it in hand. The card may jump out.

Liquid crystal screen

Normal display mode



Measurement time

Displays the elapsed time of the measurement
With [Start/Stop] button entry, time count operation is carried out until completion of the measurement time.

Display data: Displays the current value numerically.

Remaining capacity of battery.

Bar: Displays the current instantaneous level.

Display Range: Displays upper and lower limit of the bar graph displayed Displays the upper and lower limit of the range for bar graph to be displayed, selected according to the magnitude of sound level.

Cal、Pause、Card：Displays calibration, pause, and card insertion, respectively. Displays Calibration with Cal key, Pause with Pause key, and Card with card inserted, respectively.

Leq：Measurement mode.　Displays the mode selected with [Meas Mode] key.(Lp, Leq, Lmax・・・・・・・)

Fast: Time weighting characteristics: Displays the dynamic characteristics selected with　[F•S Imp] key. (Fast, Slow or Impulse)

A: Frequency weighting characteristics: Displays the frequency weighting selected with [A,C,Z] key.

Displayed when the measured value exceeds the range selected, as well as Un when the input l level is lower than the selected range.

Ov : Over　　；＋3dB from upper limited scale
Un : Under　；－0.6dB from lower limited scale

Displays operation condition y

Rec　blinking: Under the measurement started with [Start/Stop] key pushed.
Stp blinking: Measurement terminated

Measurement time

The measurement time is displayed, which is one of the following :.
1s、3s、5s、10s、1m、5m、10m、15m、30m、1h、8h、12h、24h、＊＊＊(Forever: Until Stop key entry)

Pause (Temporary interruption mark)

Blinks when the calculation or data saving to memory is canceled, where displayed level is not updated

Frequency weighting characteristics is displayed.

L_A: A - weighted

L_C: C - weighted

L_P: Z - weigjted

The third and fourth digits are displayed when each corresponding calculation is completed, implying the following : .

L_{Aeq} : Equivalent continuous A-weighted sound pressure level

L_{AE} : Single event sound exposure level

L_{Amax} : Maximum A-weighted sound pressure level

L_{Amin} : Minimum A-weighted sound pressure level

L_{A05} : Percentile (5%) sound pressure level

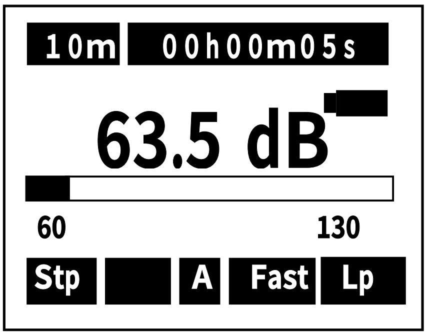
L_{A10} : Percentile (10%) sound pressure level

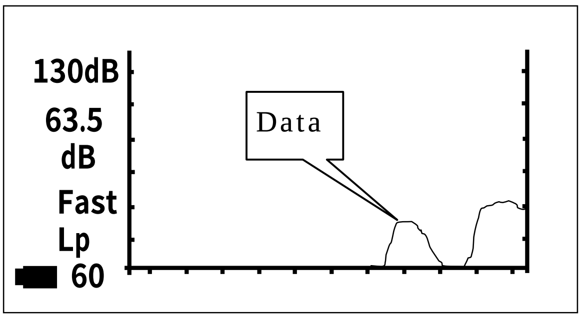
L_{A50} : Percentile (50%) sound pressure level

L_{A90} : Percentile (90%) sound pressure level

L_{A95} : Percentile (95%) sound pressure level

Example of measurement screen displayed



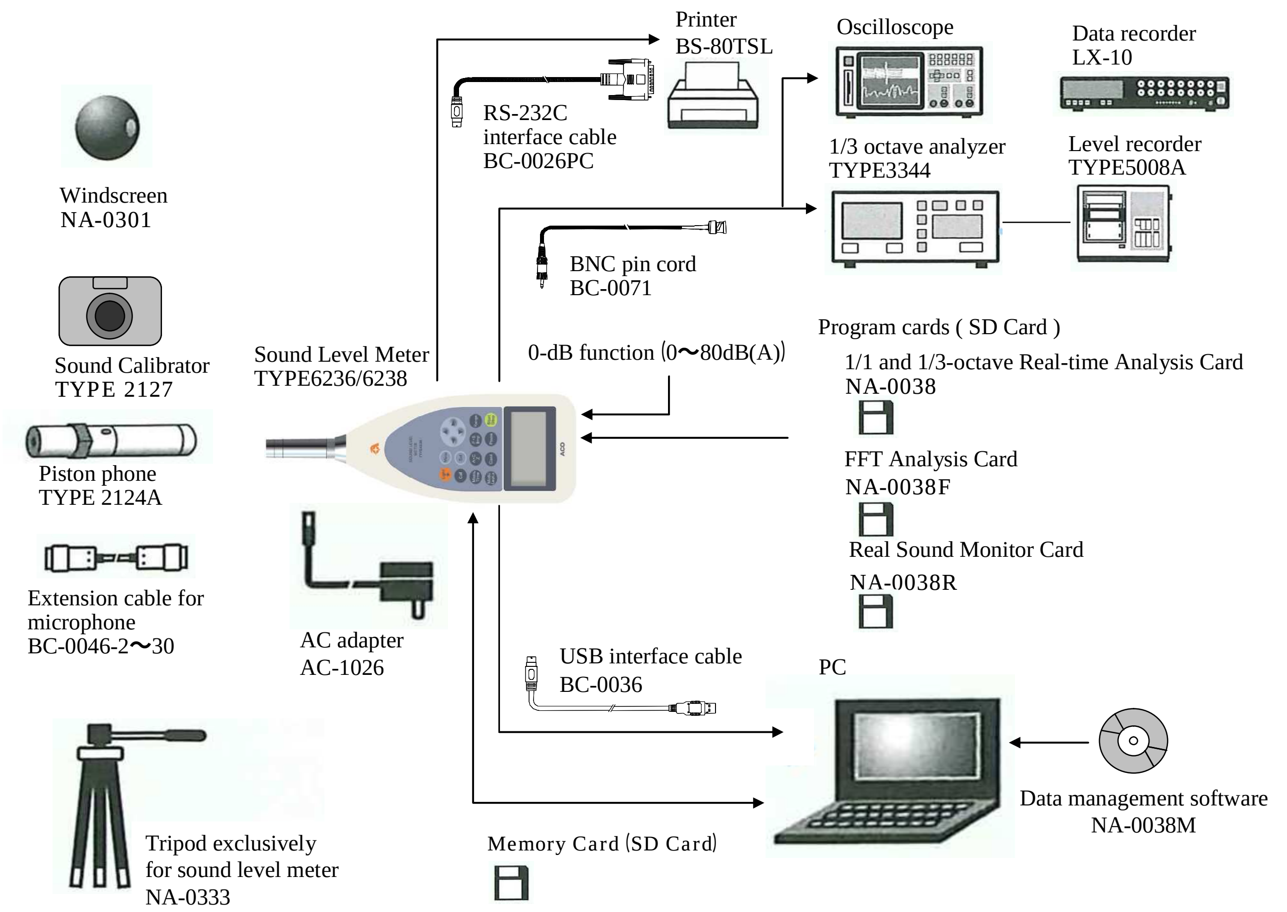


Example of T-L (time level) display

System configuation

Example of system configuration

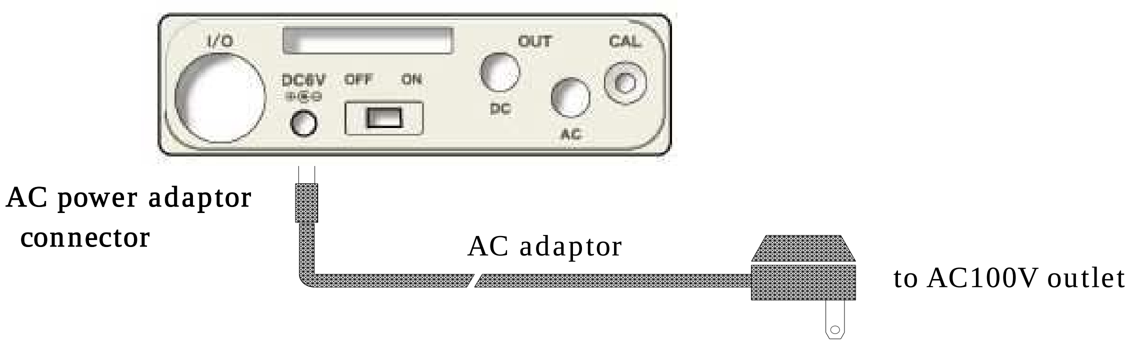
※The function can be extended by the connecting various option measuring instruments.



AC adaptor (option)

1) Turn off the power switch.
2) Connect the option AC adaptor to the AC adaptor terminal.
3) Insert the AC plug of the AC adaptor to the AC100V outlet.

NOTE
Please do not use any other power supplies than specified AC adaptor. It may cause breakdown or malfunction.



Windscreen(ϕ50)

The measurement error may be caused in the windy outdoor site or noise measurement of ventilator, since the wind drives against the microphone generating the wind noise.
Under such conditions, it is possible to reduce the wind noise by attaching the ϕ50 windscreen to the microphone

Mounting on the tripod

It is possible to mount this equipment on the camera tripod in lengthy measurement.
Please be careful enough not to drop the equipment or fell the tripod

Memory card (SD card) and program card (Option)

The measurement results can be stored in Memory card (SD card) to reedit it on personal computer
Moreover, option program cards enable to set up the conditions of 1/1 or 1/3 octave filter card, FFT analysis card, and RSR card (Real sound recording card)

Inserting and detaching the card

1. Insert the card into the card slot on the side panel.
   Press softly the card into the slot until it comes to the end, watching for the direction of the card.
   Press the card again for detaching it. The card comes off by itself.

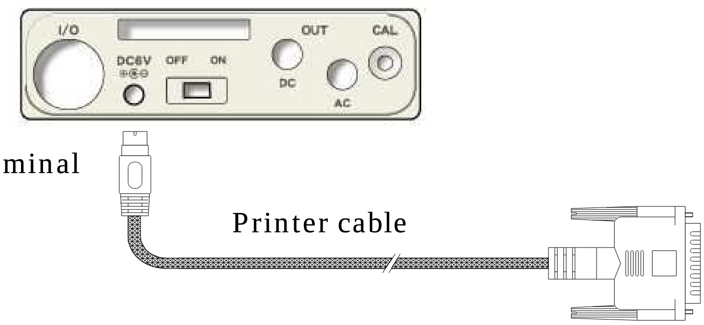
Extension cable (BC-0046)

Please make sure to switch off the power when connecting or disconnecting the microphone extension cable.
To avoid the influence of diffraction effect of the sound level meter body, or of the existence of the measuring person, microphone can be placed away from the main body.
Please refer to “3-4 Connections and How to Connect Extension cable” in P51 for further information.

NOTE
Never separate the microphone from the preamplifier, which may cause breakdown or malfunction

Connection with printer （BS2-80TS）

Connect the serial input terminal of printer (BS2-80TS) to I/O terminal on the side panel with RS-232C interface cable (BC-0026PC). (both available as option).



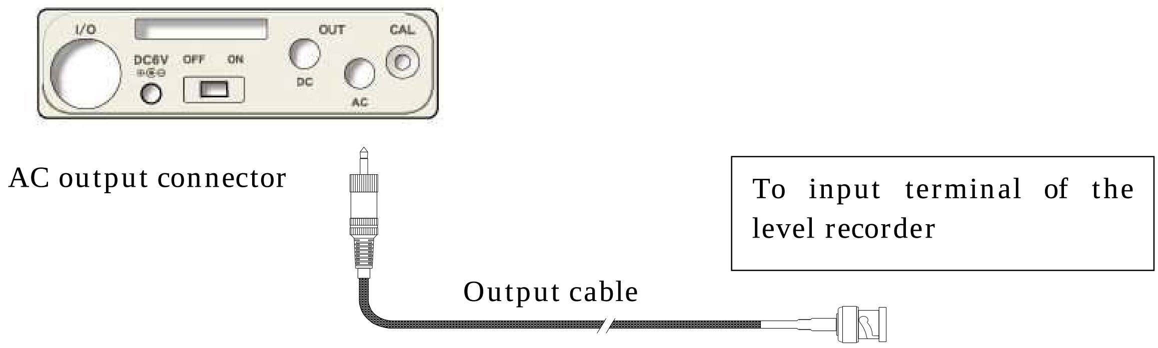
I/O terminal

To printer
To serial input terminal

Level recorder (TYPE 5008A)

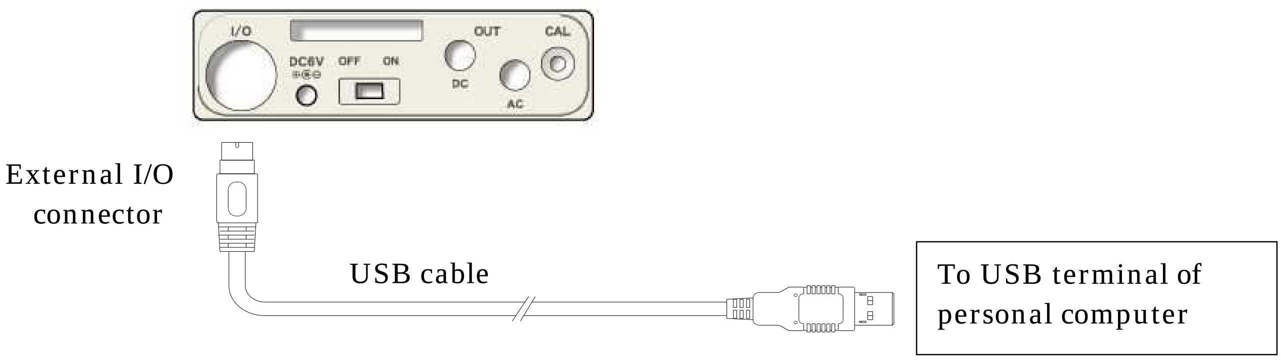
How to record the sound pressure level

Connect AC output connector on the side panel to level recorder with output cable (BC-0071 : Option) as shown in the following figure.



Connection with personal computer

External connect I/O on the side panel to USB terminal of personal computer with the option interface cable.



Preparation

Battery installation

When LCD display tells low battery, install new batteries.
For long-term measurement, install new batteries in advance.
The following displays tell you the condition of the batteries.
The battery residual quantity display are the 5 stages like the following.

It will blink, if"ENPTY" is displayed, and a power supply is shut off.

FULL LOW EMPTY Replace batteries

To install new batteries:

1) Turn off the POWER switch.
2) Push the Battery cover softly and slide it rightward.
3) Put the new batteries in the case, then shut the cover. The inside of the case shows you the direction of the batteries.

NOTE
Do not put the batteries in the wrong direction. These four batteries should be replaced at the same time.

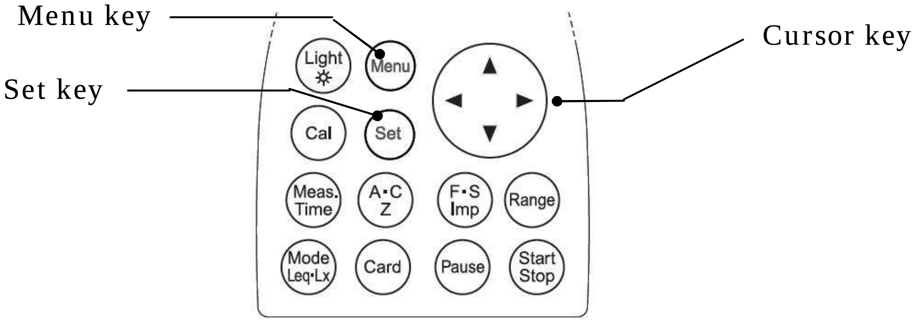
• Battery life is approximately:
  9 hours (Alkaline batteries, continuous operation)
• Use of LCD back-light shortens the life of the batteries (approximately 1/3).

NOTE
Please prepare the AC adaptor (option) in advance when it is used for a long period of time.

Calendar adjustment

To adjust the calendar (time), operate as follows.

You can adjust calendar in the Menu mode in the same way as LCD adjustment.



When you press the **[Menu]** key, the following screen appears

```
<System>                1/3
Mode            : Normal
Data delet      : off
LCD cont        : ＊＊＊
Date  y/m/d     : 00/00/00
Time   h/m/s    : 00:00:00
Printer(pc)set  : 9600
USB   out       : OFF
```

Date adjustment

Time adjustment

【Calendar adjustment】

1） Select date y/m/d with Cursor key ▼, then move the cursor rightward with ▶ key.

2） Set the year/month/day with ▲▼ key, then press [Set] key to save the setting.
After pressing [Set] key, the cursor moves to leftward.

1. If you want to go back to the measurement mode, press [set]key.

【Time adjustment】

1） Select [time] with Cursor key ▼, then move the cursor rightward with ▶ key.

2） Set the hour:minute:second with ▲▼ key, then press [Set] key to save the setting.
After pressing [Set] key, the cursor moves leftward.

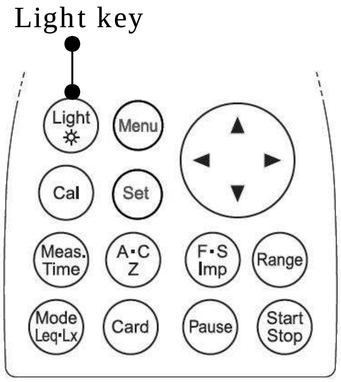
3） If you want to go back to the measurement mode, press [Set] key.

NOTE
Be sure to enter the date (date y/m/d) in the order of “year → month → day.” Input any figure of; y(year): 00 – 99, m(month):01 – 12, and d(day): 01 – 31. Ex.) For November 30, 2003 Correct) 03/11/30 Incorrect) 11/30/03 30 has been entered for m(month). Input any figure of 01 through 12. Be sure to enter the time in the order of “hour → minute → second.” Input any figure of; h(hour): 00 – 24, m(minute): 00 – 59, s (second) 00 – 59. Ex.) For 23:58:32 Correct) 23/58/32 Incorrect) 32/58/23 32 has been entered for h(hour). Input any figure of 00 through 24.

NOTE
You are recommended to set the built-in IC timer right before measurement, since it could show the wrong time.

LCD back-light

You can use LCD back-light, when your measurement is carried out in the dark situations.



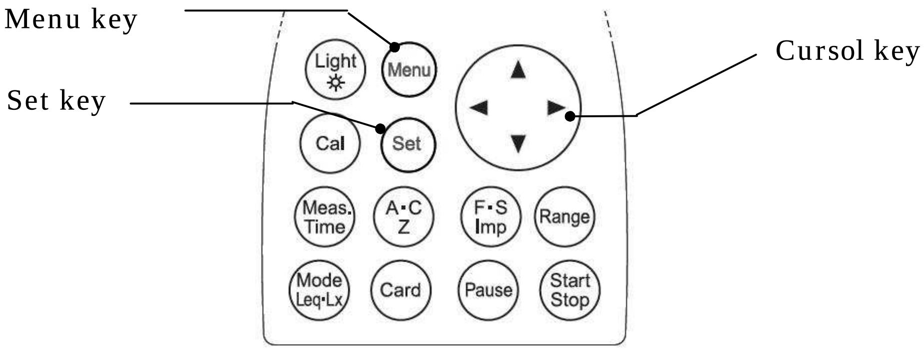
1) Press [Light] key, LCD back-light goes on.
2) If you press [Light] key again, LCD back-light goes out.
   The light automatically goes out in about 30 seconds after the light goes on.
3) When the batteries is low, LCD back-light dims.

NOTE
Use of LCD back-light shortens the life of the batteries.

LCD adjustment

You can adjust LCD contrast, when the batteries were low, or when the new batteries were installed.

The procedure is as follows.



1) When you press the [Menu] key, the following screen appears.

```
<System>                 1/3
Mode            : Normal
Data delet      : off
LCD cont        : ＊＊＊
Date  y/m/d     : 00/00/00
Time h/m/s      : 00:00:00
Printer(pc)set  : 9600
USB   out       : OFF
```

LCD adjustment

2) Select LCD cont with Cursor key ▼, then move the cursor rightward with ▶ key.

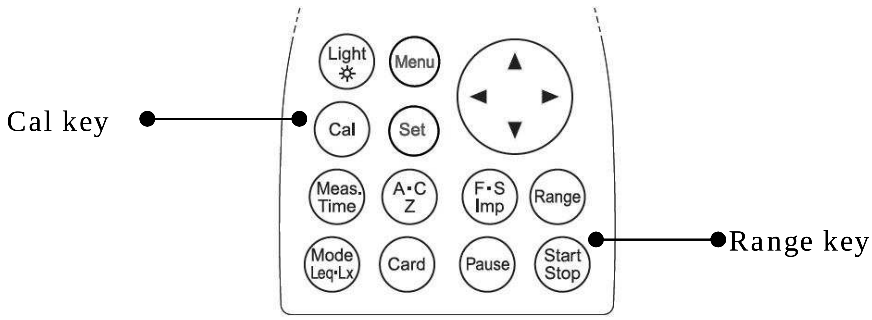
3) Adjust the LCD contrast with ▲▼ key, then press [Set]key to save the setting.
After pressing [Set] key, the cursor moves to leftward.
If you want to go back to measurement mode, press [Set] key.

Calibration

You need to calibrate the instrument regularly before you start taking measurements. There are two types of calibration. One is the way using the internal generator, the other is the way using the pistonphone. Note that calibration is disabled when “Peak measurement” is selected

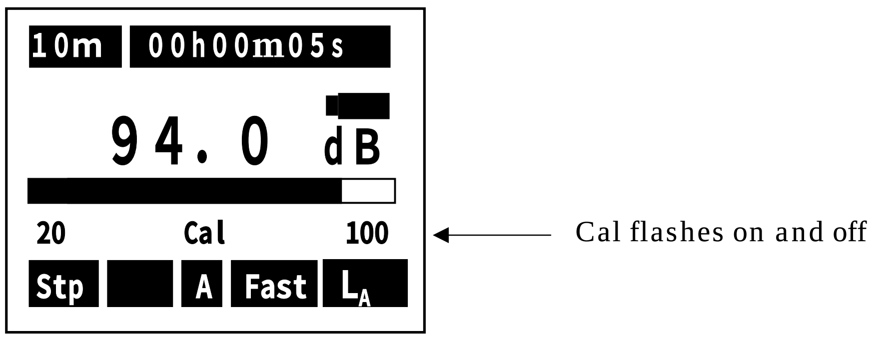
Calibration using internal generator

You can calibrate the instrument using the internal generator (1kHz, sine wave)

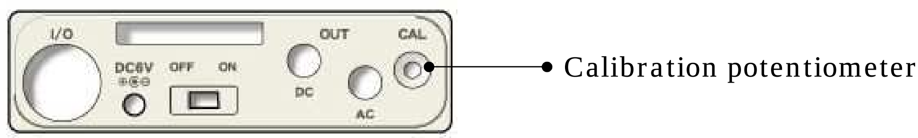


1) Turn on the POWER.
2) Press **Cal** key.
3) Press Range key, and choose ‘100dB’ by cursor keys ▲▼, and press Range key again to register.
4) Adjust the calibration potentiometer on the side panel until the display shows 94dB.
5) If **Cal** key is pressed once again, the calibration is completed.

< Calibration display >



< Side panel >



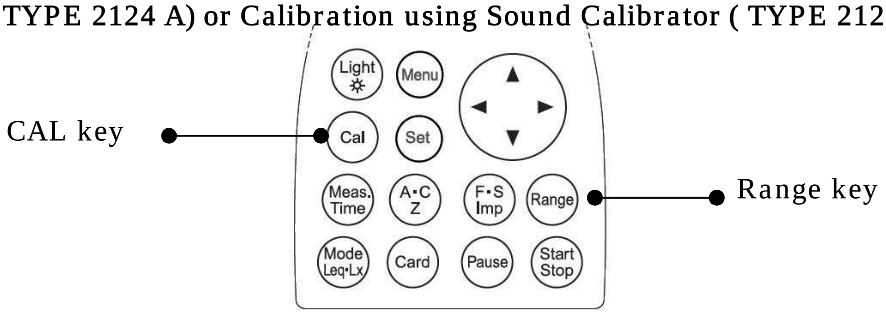
< Reference > Full scale range and Cal(the display shows)

Full scale range (dB)	CAL (dB)	OUTPUT (V)	
		AC OUT	DC OUT
80	74.0	0.500	2,350
90	84.0	0.500	2,350
100	94.0	0.500	2,350
110	104.0	0.500	2,350
120	114.0	0.500	2,350
130	124.0	0.500	2,350

<**Reference**> Relation between the display value of each range, and output voltage

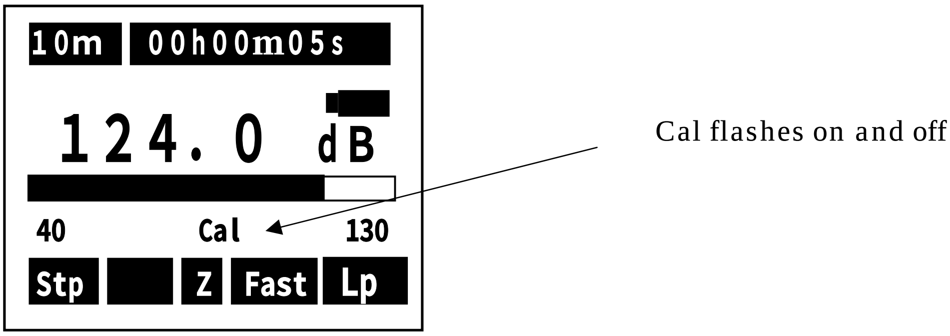
DISPLAY VALUE (dB)						OUTPUT VOLTAGE (V)	
RANGE						AC OUT	DC OUT
40～130	30～120	20～110	20～100	20～90	20～80		
130	120	110	100	90	80	1.00000	2.50000
120	110	100	90	80	70	0.31623	2.25000
110	100	90	80	70	60	0.10000	2.00000
100	90	80	70	60	50	0.03162	1.75000
90	80	70	60	50	40	0.01000	1.50000
80	70	60	50	40	30	0.00316	1.25000
70	60	50	40	30	20	0.00100	1.00000
60	50	40	30	20	—	0.00032	0.75000
50	40	30	20	—	—	0.00010	0.50000
40	30	20	—	—	—	0.00003	0.25000

Pistonphone (TYPE 2124 A) or Calibration using Sound Calibrator (TYPE 2127)



1) Turn off the POWER of Pistonphone (TYPE 2124A) or Sound Calibrator (TYPE 2127).
2) Turn on the POWER of this equipment
3) Set the frequency weighting to Z with **Frequency weighting** key, set the time weighting to **Fast** with **Time weighting** key and set the range to **40～130dB** with **Range** key.
4) Insert microphone of this equipment to Pistonphone (TYPE 2124A) or Sound Calibrator (TYPE 2127).
5) Switch on the Pistonphone (TYPE 2124A) or Sound Calibrator (TYPE 2127).
6) Adjust the calibration potentiometer on the side panel until the display shows a output level of the pistonphone (standard value is 124dB) and a output level of the sound Calibration (standard value is 94dB).

< **Calibration display** >

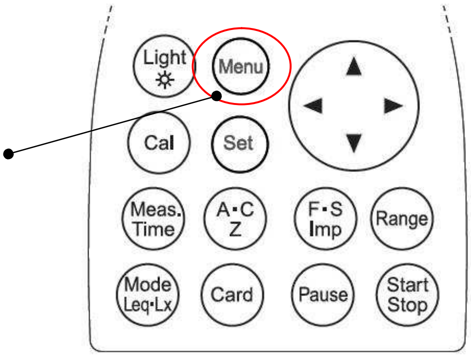


NOTE
Insertion and extraction of the microphone to/from the sound calibrator should be made slowly and softly. Rapid insertion and extraction may cause strong force to the diaphragm of the microphone due the air pressure change, which may then give a severe damage to the microphone.

Menu screen

How to use Menu

Menu key



With [Menu] button pressed, the following Menu screen appears. (Under the situation with [Start] key not pressed).

[Menu 1/3] → [Menu 2/3] → [Menu 3/3] → [Menu 1/3] → [Menu 2/3]・・・

Each [Menu] key pressed, you can select one of three screens as above, and return to the measurement setting screen. .

Select an item with cursor keys ▲ ▼, start the input with ▶, and fix the change with [Set] key. Move to the item to change and return to the measurement setting screen with [Set] key again.

In the use of option card (filter card), additional setting screen appears.

Please refer to the manual of each option card for the details .

```
<System>            1/3
Mode           : Normal
Data delet     : off
LCD cont       : ＊＊＊
Date y/m/d     : 01/01/01
Time h/m/s     : 00:00:00
Printer(pc)set : 9600
USB  out       : OFF
```

```
<Memory>            2/3
Mode           : Normal
Interval       : Single
```

```
<View Mode>          3/3
Lp    : INST
(A)View        LA05 : OFF
LAeq  : ON     LA10 : OFF
LAE   : OFF    LA50 : OFF
LAmin : ON     LA90 : OFF
LAmax : ON     LA95 : OFF
```

Each change made with various key operations are registered and reproduced on next Power ON operation.

<System> (1/3)

```
<System>            1/3
Mode          : Normal
Data delet    : off
LCD cont      : ＊＊＊
Date  y/m/d   : 01/01/01
Time h/m/s    : 00:00:00
Printer(pc)set : 9600
USB   out     : OFF
```

Item	Default	Contents
● Meas Mode	：Normal	：Normal ：Normal measurement Print ：Print PC out ：Data management Memory Call ：Display recorded data L_{Atm5} ：Tact-max sound pressure level (STD card) Remote ： Communicate mode
● Data delet	：Off	：Data deletion mode setting Off ：Date deletion is disabled Fixed to OFF in Peak mode 3sec ：Data in past 3 sec is deleted when [**Pause**] key is pressed during the measurement. 5sec ：Data in past 5 sec is deleted when [**Pause**] key is pressed during the measurement . ※The function is disabled for **Meas Time** 1, 3 or 5 sec,.
● LCD cont	： *****	：Adjustment of LCD contrast See "LCD adjustment" section for the details.
● Date y/m/d	：00/01/01	：Calendar setting (date: 2000/01/01) See "Calendar adjustment" for the details.
● Time h/m/s	： 00:00:00	：Time setting See the "Calendar adjustment" for the details.
● Printer(PC) set	： 9600	：Baud rate setting (9600 or 19200)
● USB out		：Digital data output setting OFF → L_p → L_{pB} → Wave (Outputs data from USB out in parallel with the measurement.) OFF ；USB output is disabled. L_p ；Outputs instantaneous value in each second L_{pB} ；Outputs level data in each band when the octave filter is used. Wave ；Outputs level data in each band at sampling rate 48kHz when the octave filter is used.

<Memory>(2/3)

<Memory> 2/3
Mode : Normal
Interval : Single

Select to
Mode:Auto

<Memory> 2/3
Mode : Auto
Interval : Single
I/O : OFF
Level : 65dB
Samp Time : Meas Time
Sta : 08/10/10 18:16:00
Stp : 08/10/12 20:16:00

Items	Default	Explanation
● Mode	: Normal	:Normal Basic setting Auto Automatic measurement, where the following items are available.
● Interval	: Single	:Measuring interval setting Single : The measurement starts with **[Start]** key and is terminated at **Meas Time** selected . Repeat : The measurement starts with **[Start]** key and is repeated in every **Meas Time** selected until **[Stop]** key is pressed.
● I/O	: OFF	:External output setting OFF : Default (Data output is disabled). ON : Outputs data for one second when the data mory mode is active.
● Level	: 65dB	: 65dB : Threshold level is registered (when the level exceeds it, recording starts), within the range 20-130dB at resolution 1dB
● Samp Time	: Meas Time	: Meas Time : sampled at interval equal to Meas time. 100ms : sampled at interval 100ms (0.1s). 200ms : sampled at interval 200ms (0.2s).

** Meas Time is time set with [Meas Time] key (1s〜-•••).

Fixed to Mease Time, when when RSR card is installed.

＊ Select 10s or more in L_{Atm5} measurement.

● Sta : Registers the starting time for recording (YY/MM/DD HH/MM/SS) (Year/Month/Date, date time/minute/second).

● Stp : Registers the stop time for recording (YY/MM/DD HH/MM/SS) (Year/Month/Date, date time/minute/second)

.

NOTE
Measurement starts when the selected level is exceeded after the time specified with [Sta Time], In the following example : : When the level exceeds 65dB after 18:16 October 10, Recording starts and the measurement is made once during the time specified with [Meas Time]. Recording is continued, in Interval Repeat mode, until the level falls or until 20:16 October 12,. Data is shown according to System Mode.

<View Mode> 3/3

Select the category of displayed data.
The data registered here is displayed in the standard screen, one by one with [Mode] key pressed on the main body.

```
<View Mode>          3/3
Lp     : INST
(A)View          LA05 : OFF
LAeq  : ON       LA10 : OFF
LAE   : OFF      LA50 : OFF
LAmin : ON       LA90 : OFF
LAmax : ON       LA95 : OFF
```

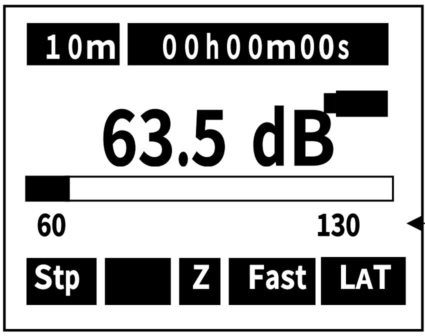
Lp : Instantaneous value
Leq : Equivalent continuous A-weighted sound pressure level
L_E : Single event sound exposure level
Lmin : Minimum A-weighted sound pressure level
Lmax : Maximum A-weighted sound pressure level
Lpeak : Peak sound pressure level

Items	Default	Explanation	
● L	: INST	: INST	: Data is displayed in each second
		TACT	: The maximum level is displayed in one second. (TACT MAX)

In each [Mode] key pushed in the measurement screen, display changes as follows :

- L_A→Leq→L_E→Lmax→Lmin→Peak→L_{05}→L_{10}→L_{50}→L_{90}→L_{95}
- When TACT is selected for Lp, [LaT] is displayed in display mode.

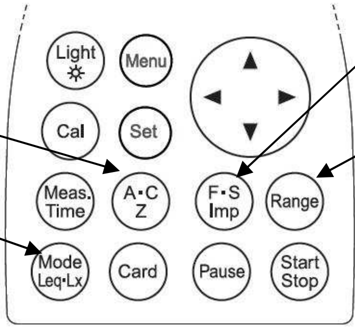
<Display> When Lp is TACT



Display mode
In the case of Lp : TACT
Z : L_PT
A : L_AT
C : L_CT

Measurement Procedure

Sound pressure level (L_p) measurement：Frequency weighting key Z

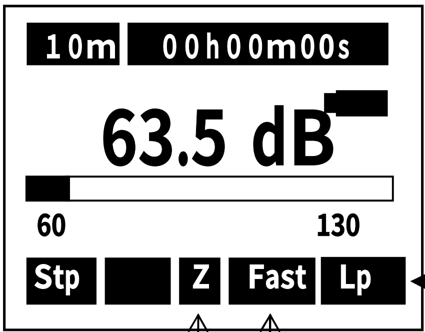


＜ Parameter setting ＞

Measure ment is made according to the following procedure.

①Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：F, S or Imp
③Frequency weighting key	：Z
④Display mode key	：L_p

＜ Display ＞

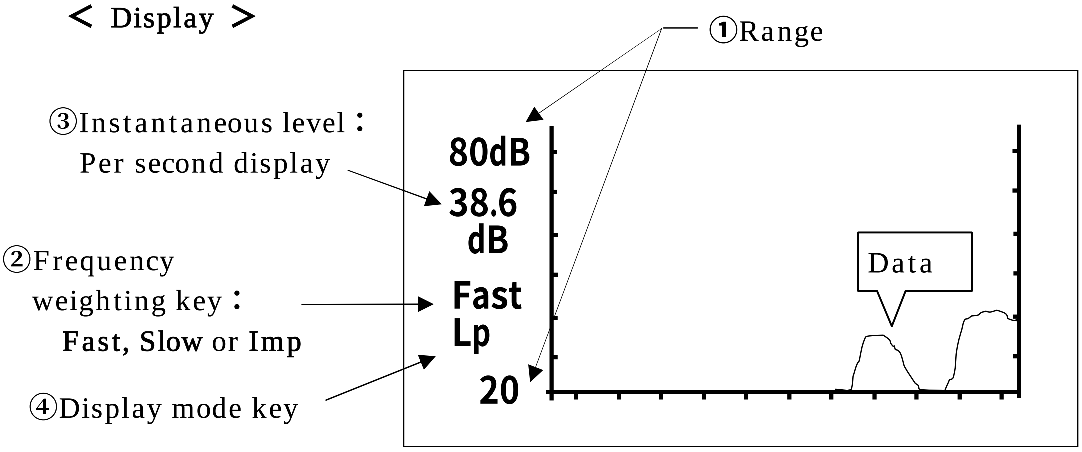


Temporal level display of Sound pressure level (Lp)

＜ Parameter setting ＞:

The temporal level is displayed at each contiguous push (1.5s) of Mode key as follows, returning to the standard display screen when the key is pushed again.
The key operation is similar to the measurement of sound pressure level (Lp).

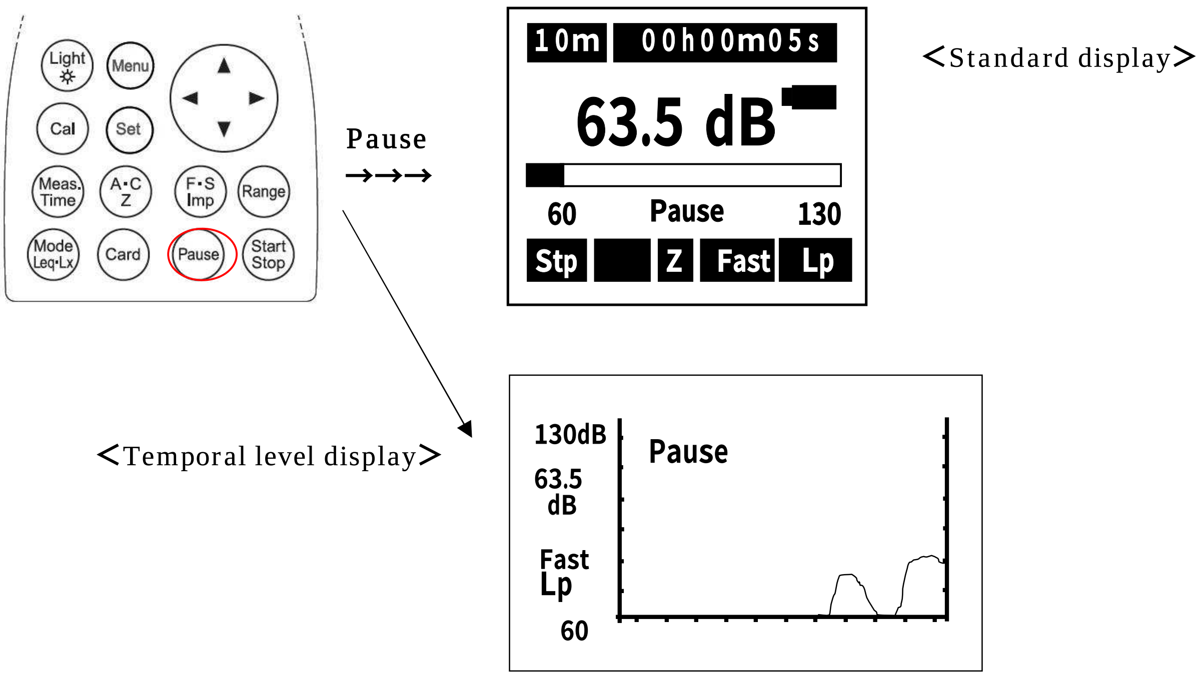
＜ Display ＞



The instantaneous level is displayed at each about 300ms from right to left.

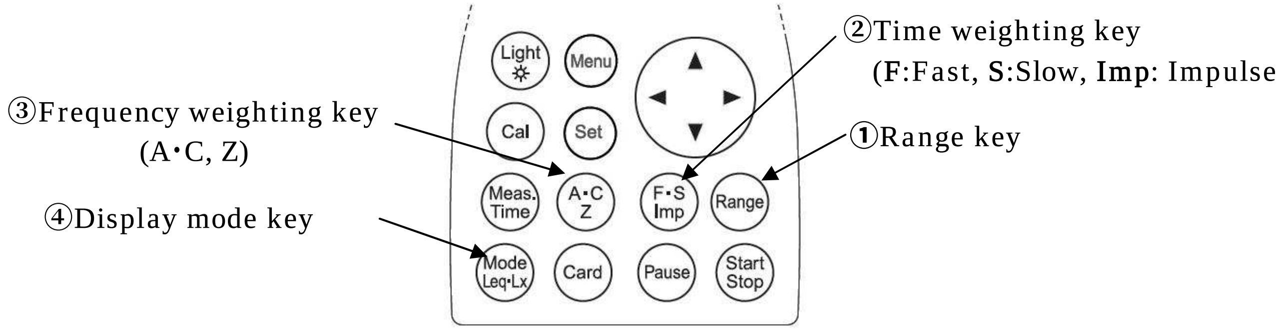
Data hold

By pushing the **Pause** key, the blinking letter “Pause" is displayed at the center of the bar graph, displaying the present instantaneous level. Note that the bar graph itself doesn't pause.



• By pushing the Pause key is pushed again, it is released.

A-weighted sound pressure level (L_A/L_C) measurement：Frequency weighting key A, C

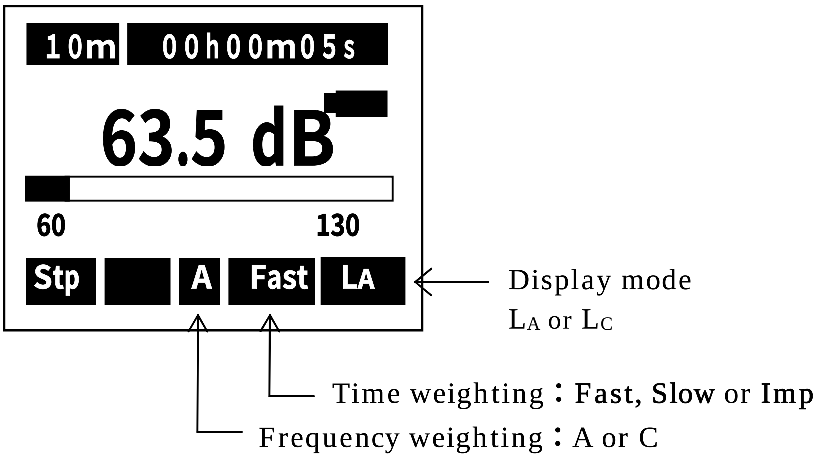


＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：F, S or Imp
③Frequency weighting key	：A or C
④Display mode key	：L_A , L_C

＜ Display ＞

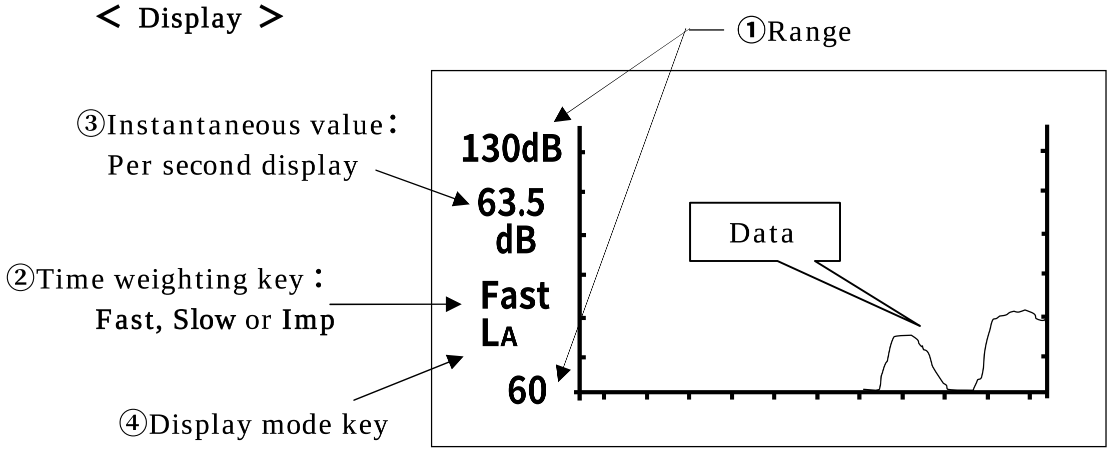


Time level display of A-weighted sound pressure level (LA/LC)

< Parameter setting >

The key operation is similar to the measurement of A-weighted sound pressure level (LA/LC).

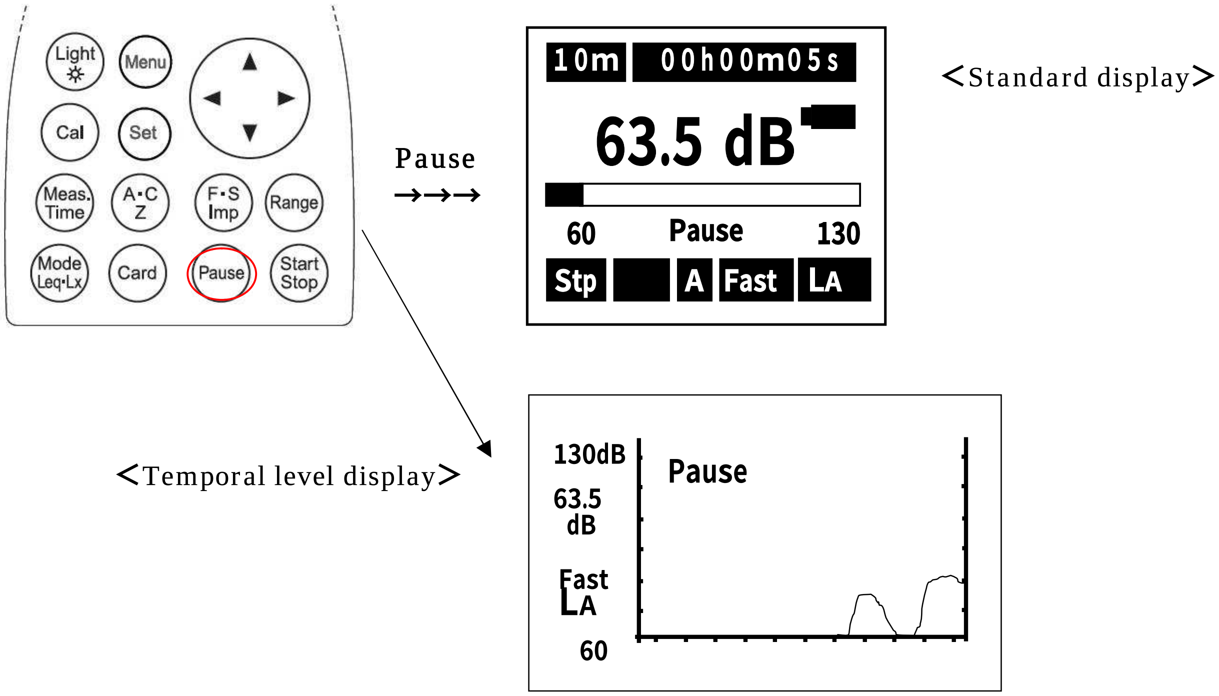
< Display >



The instantaneous level is displayed at each about 300ms from right to left.

Data hold

By pushing the **Pause** key, the blinking letter "Pause" is displayed at the center of the bar graph, displaying the present instantaneous level. Note that the bar graph itself doesn't pause.



• By pushing the [Pause] key again, it is released.

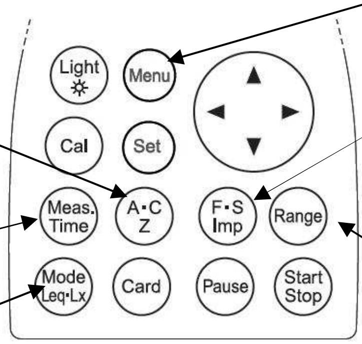
Equivalent continuous A-weighted sound pressure level (L_{Aeq}) measurement

2) Menu key

③ Frequency weighting key (A・C, Z)

②Time weighting key (F:Fast, S:Slow)

④Measuring time key



⑤Display mode key

①Range key

＜Menu＞

<View Mode> 3/3
Lp : INST
(A)View LA05 : OFF
LAeq : ON LA10 : OFF
LAE : OFF LA50 : OFF
LAmin : OFF LA90 : OFF
LAmax : OFF LA95 : OFF

＜ Parameter setting ＞

1) The key operation is similar to the measurement of A-weighted sound pressure level (L_A) except that it needs [Start/Stop] key input for starting the measurement (automatic calculation).

2) To display the value L_{eq}, keep the "LAeq" key ON in advance in the Menu mode.

① Range key：Select a range where the bar graph indicates approximately 2/3 of the full scale.

②Time weighting key ：F or S

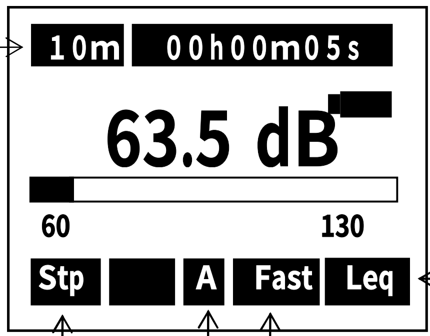
③Frequency weighting key ：A

④Measurement time key ：**1s、3s、5s、10s、1m、5m、10m、15m、30m、1h、8h、24h** and ＊＊＊(until the Start/**Stop** key pushed again)

⑤Display mode key ：L_{eq}

＜ Display ＞

Measurement



Display mode Leq

Rec:Blinking during recording

Stp: Holding state

Time weigh

influence the measurement.)

Frequency weighting：**A**

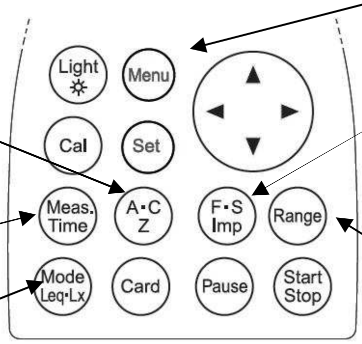
- Measurement starts with **Start** key pushed, and ends up automatically at the measurement time. Digital display indicates the halfway result at the current point of time. (Display " Rec " blinks while the measurement.)
- When **Interval** is set to **Repeat** in <Menu> mode, the measurement is repeated in every measurement time. (This is used when continuous measurement is needed.)
- By pushing **Stop** key in course of the measurement, calculation is done using the data so far.
- By pushing **Pause** key in course of the measurement, the calculation can be done without using the data in the latest 3 or 5 seconds.
- To set this function, see the description of **data delete** in "Section 4 Menu 2.Menu(1/2) ". This function can be set in the Data delete in the Menu mode.
- When *** is selected, the final data is calculated and displayed only when **Stop** is pushed or 199 hours have gone through.
- All the keys do not respond during the measurement : Start／Stop、Mode、Light

Single event sound exposure level (L_{AE}) measurement

③Frequency weighting key (A･C, Z)

④Measuring time key

⑤Display mode key



2) Menu key

②Time weighting key (F:Fast, S:Slow, Imp: I mpulse)

①Range key

＜Menu＞

<View Mode> 3/3
Lp : INST
(A)View LA05 : OFF
LAeq : OFF LA10 : OFF
LAE : ON LA50 : OFF
LAmin : OFF LA90 : OFF
LAmax : OFF LA95 : OFF

＜ Parameter setting ＞

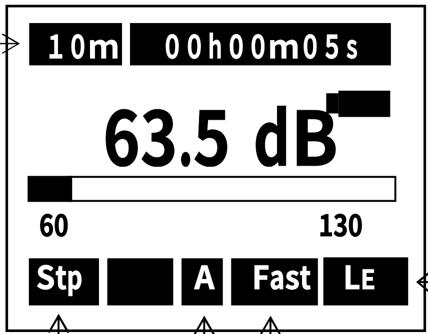
1) The key operation is similar to the measurement of A-weighted sound pressure level (L_A) except that it needs **[Start/Stop]** key input for starting the measurement (automatic calculation).

2) To display measurements of L_E, input ON to "LAE" beforehand by the Menu item.

①Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：Any of F, S or Imp (doesn't influence the measurement.)
③Frequency weighting key	：A
④Measurement time key	：1s、3s、5s、10s、1m、5m、10m、15m、30m、1h、8h、24h and ＊＊＊(to the Start/Stop key)
⑤Display mode key	：L_E

＜ Display ＞

Measuring time



Display mode L_E

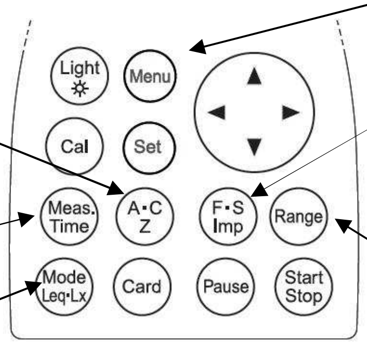
Rec:Blinks during recording
Stp: Holding state

Time weighting：Any of F, S or Imp (doesn't influence the measurement.)

Frequency weighting ：A

・The measurement is similar to the equivalent continuous A-weighted sound pressure level (L_{Aeq}).

Maximum A-weighted Sound pressure Level (L_{Amax}/L_{Amin}) Measurement



<Menu>

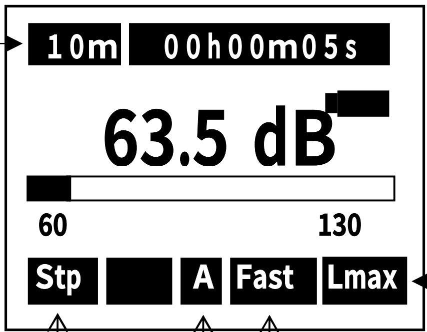
<View Mode>		3/3
Lp :INST		
(A)View	LA05	: OFF
LAeq : OFF	LA10	: OFF
LAE : OFF	LA50	: OFF
LAmin : ON	LA90	: OFF
LAmax : ON	LA95	: OFF

＜ Parameter setting ＞

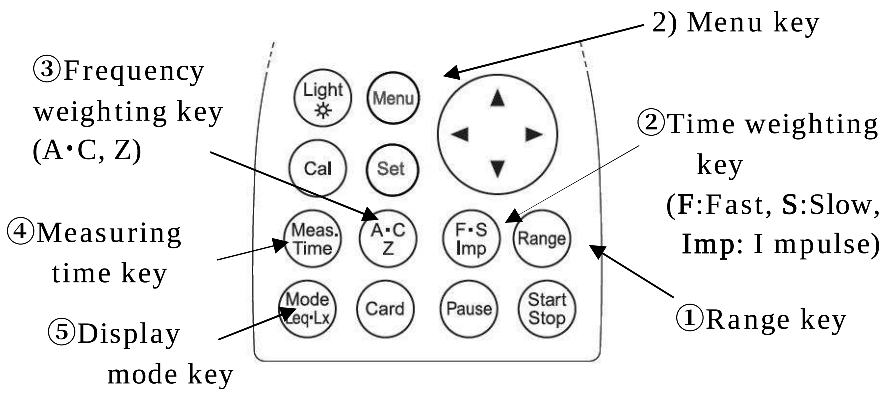
1）The key operation is similar to the measurement of A-weighted sound pressure level (L_{Aeq})
To display measurements of L_{max}, input ON to "LAmax" beforehand by the Menu item.
(similar in L_{min} measurement.)

①Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：Fast or Slow
③Frequency weighting key	：A
④Measurement time key	：1s、3s、5s、10s，1m，5m，10m，15m、30m，1h、8h、24h and＊＊＊(to the Stop key)
⑤Display mode key	：L_{max} or L_{min}

＜ Display ＞



Percentile level (L_{AN}) measurement



<Menu>

```
<View Mode>          3/3
Lp     :INST
(A)View        LA05  : ON
LAeq  : OFF    LA10  : ON
LAE   : OFF    LA50  : ON
LAmin : OFF    LA90  : ON
LAmax : OFF    LA95  : ON
```

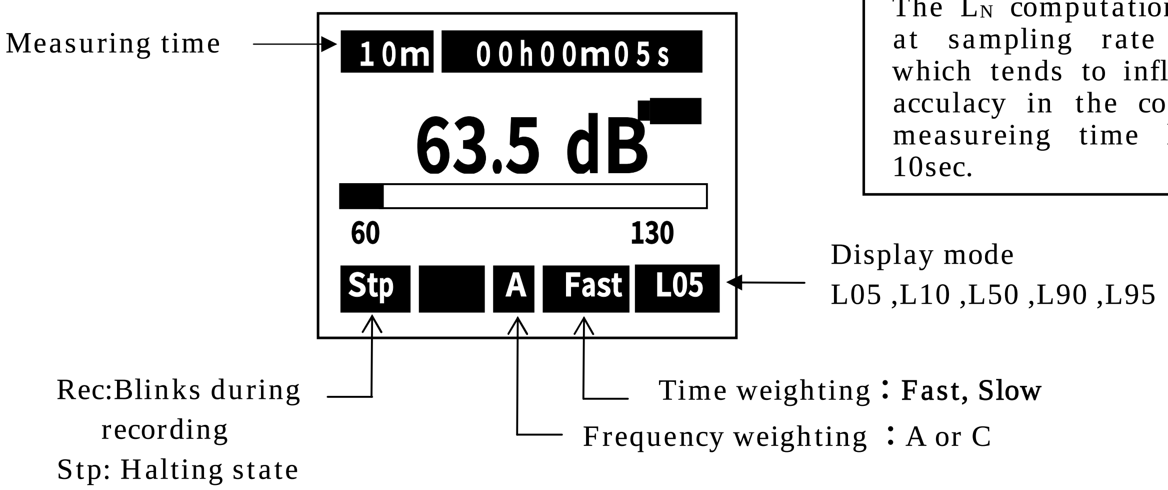
< Parameter setting >

1) The key operation is similar to the measurement of A-weighted sound pressure level (LAeq)

2) To display the value **Lmax**, keep the "LA05 ,LA10 ,LA50 ,LA90 ,LA95" key ON in advance in the Menu mode.

①Range key	: Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	: Fast or Slow
③Frequency weighting key	: A
④Measurement time key	: 1s、3s、5s、10s，1m，5m，10m、15m、30m，1h、8h、24h and＊＊＊(input to the Stop key)
⑤Display mode key	: L_N (To display the Percentile level (L_N) Measurement.)

< Display >



NOTE
The L_N computation is made at sampling rate 100msec, which tends to influence the acculacy in the condition of measureing time less than 10sec.

Additional index (L_{peak}, L_{Cpeak}, L_{Ceq}, L_{Atm5}, L_{AI}, L_{AIeq}) mesurement

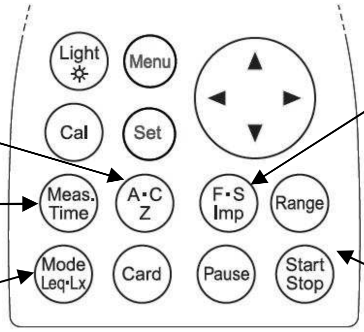
In addition to L_{eq}, L_E, L_{max}, L_{min} and L_N, the following calculations can simultaneously be made:

- L_{peak} : peak sound pressure level
- L_{Cpeak} : C-weighted peak sound pressure level
- L_{Ceq} : C-weighted equivalent continuous sound pressure level
- L_{Atm5} : Power average of maximum sound pressure level in a given interval
- L_{AI} : Impulse sound pressure level
- L_{AIeq} : Impulse equivalent continuous sound pressure level

③Frequency weighting key
(A･C, Z)

④Measurement time

⑤Display mode key



②Time weighting key
(**F**:Fast, **S**:Slow, **Imp**: Impulse)

①Range key

Peak sound pressure level (L_{peak}) measurment

The peak sound pressure level is peak sound pressure level of the sound wave before smoothed with the time weighting characteristics.
L_{peak} is peak sound level for Z characteristics.

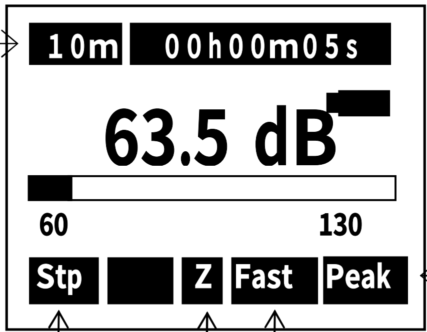
＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key ：Select a range where the bar graph indicates approximately 2/3 of the full scale.
② Time weighting key ：F, S or Imp
③ Frequency weighting key ：Z
④ Measurement time key ：1s、3s、5s、10s，1m，5m，10m，15m、30m，1h、8h、24h and＊＊＊(input to the Stop key)
⑤ Display mode key ：Peak

＜ Display ＞

Measuring time



Display mode
Peak

Rec:Blinks during recording
Stp: Holding state

Time weighting：Fast, Slow, Imp

Frequency weighting ：Z

C-weighted peak sound pressure level (L_{Cpeak}) measurement

The peak sound level is peak sound pressure level before smoothed with the time weighting characteristics.
Lcpeak is wavy peak level of C characteristic.

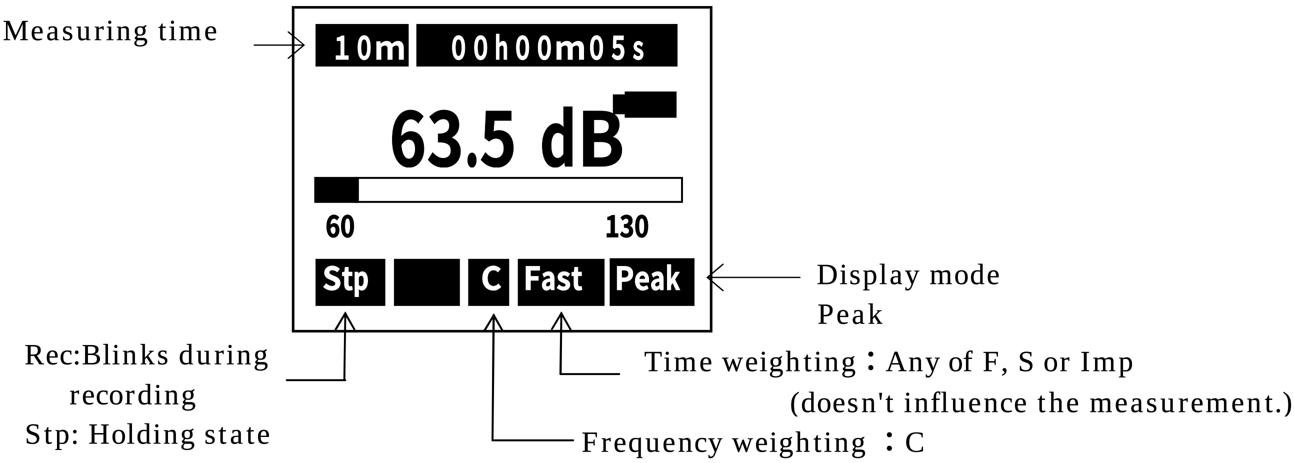
＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
② Time weighting key	：Any of F, S or Imp (doesn't influence the measurement.)
③ Frequency weighting key	：C
④ Measurement time key	：1s、3s、5s、10s，1m，5m，10m，15m、30m，1h、8h、24h and＊＊＊(input to the Stop key)
⑤ Display mode key	：Fixed to Peak

The measurement starts with [Start] key.

＜ Display ＞



- Measurement starts with **[Start]** key pushed, and ends up automatically at the measurement time. Digital display indicates the halfway result at the current point of time. (Display “ Rec ” Blinks while the measurement.)
- By pushing **[Stop]** key in course of the measurement, calculation is done using the data so far.
- When *** is selected, the final data is calculated and displayed only when [Stop]is pushed or 199 hours have gone through.

C-weighted equivalent continuous sound pressure level (L_{Ceq}) measurement

Lceq is C-weighted equivalent continuous sound pressure level.

＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key ：Select a range where the bar graph indicates approximately 2/3 of the full scale.

② Time weighting key ：**F or S**

③ Frequency weighting key ：C

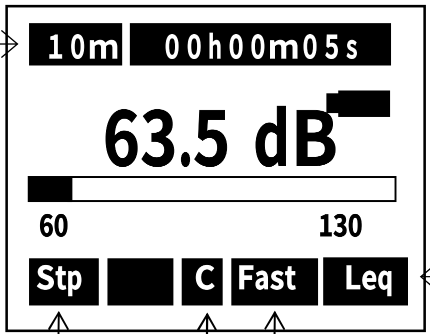
④ Measurement time key ：**1s、3s、5s、10s，1m，5m，10m，15m、30m，1h、8h、24h and**＊＊＊(input to the **Stop key**)

⑤ Display mode key ：Leq

The measurement starts with [Start] key.

＜ Display ＞

Measuring time



Display mode Leq

Rec: Blinks during recording

Stp: Holding state

Time weighting： F or S

Frequency weighting ：C

Power average value of the maximum sound pressure level in a given interval (L_{Atm5}) measurement

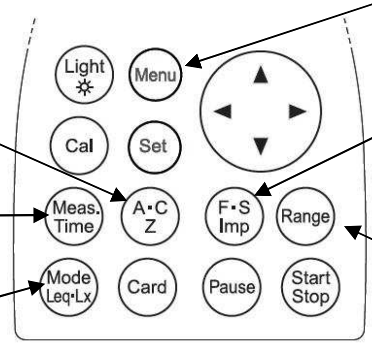
Power average value of the maximum sound pressure level in a given interval (L_{Atm5}) is power average of the maximum value of A-weighted sound pressure level in successive 5-sec intervals.
It can be measured only when A characteristics is selected in the standard screen.

1) Menu key

③Frequency Weighting key

④Measurement key

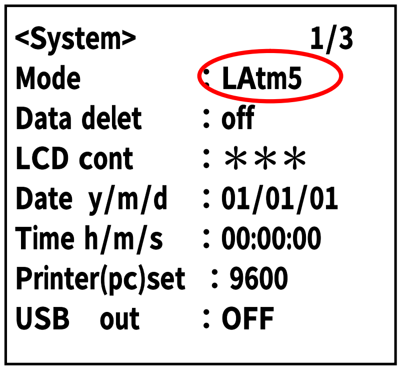
⑤Display mode



②Time weighting key (F:Fast, S:Slow, Imp: I mpulse)

① Range key

<Menu>



< Parameter setting >

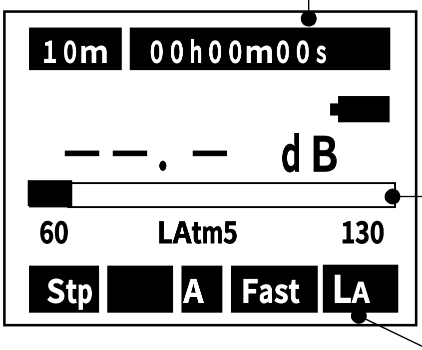
Mode: Normal in < System>1/3 is changed to Mode:L_{Atm5} with the ▲▼key in Menu screen. The screen for power average of the maximum value appears when the change is fixed and resisted with Set key.

Measurement is mede according to the following procedure

① Range key	:	Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	:	F or S
③ Frequency weighting key	:	A
④ Measurement time key	:	1s、3s、5s、10s，1m，5m，10m，15m、30m，1h、8h、24h and＊＊＊(input to the Stop key)
⑤ Display mode key	:	L_A or tm5

The measurement starts with [Start] key.

< Display >



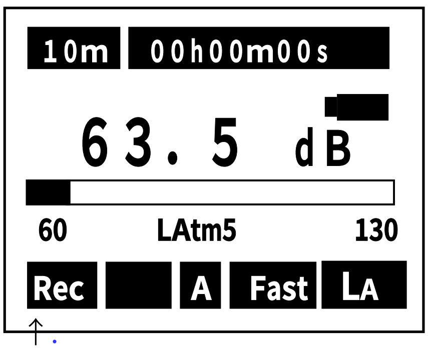
Measurement time 10s (second) display
00h00m00s
00h00m01s
00h00m02s
・
00h00m09s ← End in ten seconds
00h00m00s ← The following measurement

＊When Menu Intr:Single is measured, it becomes Repeat continuous work as the measurement time display.

Bar: Displays the current instantaneous level.

L_A or tm5

Press ［Start/Stop Set］ key



← Count-up timer for the measurement

← Starts with display ――.― when the measurement mode is tm5 .
Measurement value is displayed every five second.

← The bar displays the instantaneous level in every 0.1 second.

← L_A and tm5 can be changed.
Current level is displayed with L_A selected value.

↑ The Rec blinks with [Start] key input

Impulse sound pressure level(L_{AI}) measurement

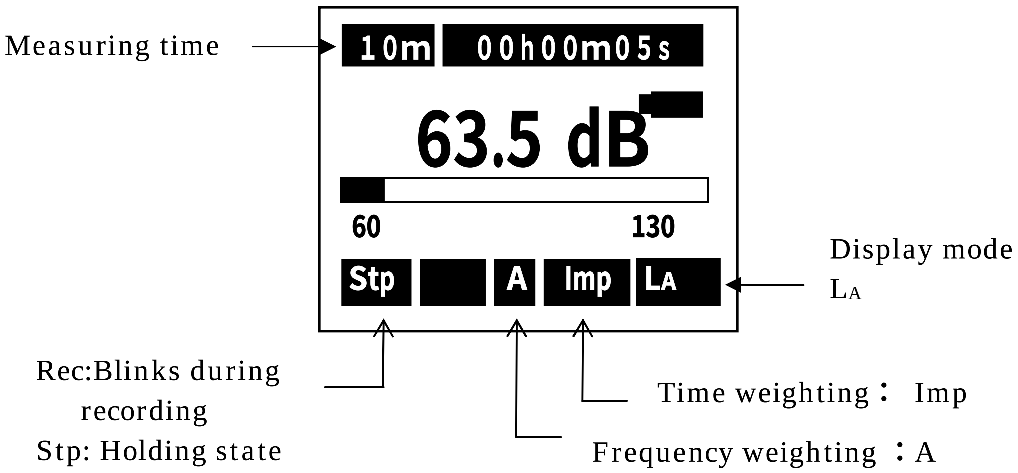
Impulse sound pressure level (L_{AI}) is A-weighted sound pressure level with time weighting characteristics, 'Impulse' .
It can be used only when A characteristics is selected in the default screen.

＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key	：Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：Imp
③ Frequency weighting key	：A
④ Display mode key	：L_A

＜ Display＞



Impulse equivalent continuous A-weighted sound pressure level (L_{AIeq}) measurement

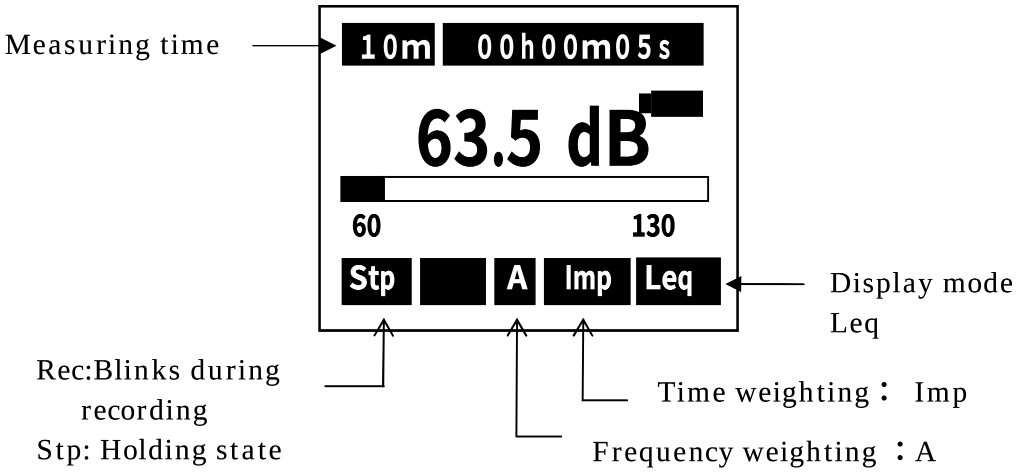
Impulse equivalent continuous A-weighted sound pressure level (L_{AIeq}) is equivalent continuous sound pressure level with time weighting characteristics, ‘Impulse’.
It can be used only when A characteristics is selected in the default screen.

＜ Parameter setting ＞

Measurement is mede according to the following procedure

① Range key	：	Select a range where the bar graph indicates approximately 2/3 of the full scale.
②Time weighting key	：	Imp
③ Frequency weighting key	：	A
④ Display mode key	：	Leq

＜ Display＞



Memory (AUTO)

＜ Operation ＞

By changing Mode: Normal in < Memory>2/3 of [Menu] screen to Mode: Auto with the ▲▼key , and fixing it with [Set] key, the following screen appears:

```
<Memory>          2/3
Mode        : Normal
Interval    : Single
```

Select to
Mode：Auto

```
<Memory>            2/3
Mode          : Auto
Interval      : Single
I/O           : OFF
Level         : 65dB
Samp Time     : Meas Time
Sta : 08/10/10    18:16:00
Stp : 08/10/12    20:16:00
```

- I/O ： OFF ：External output setting
  - OFF ： Default (Data output is disabled).
  - ON ： Outputs data for one second when the data mory mode is active.
- Level ：Threshold level is registered (when the levelexceeds it, recording starts)
- Samp Time ：Meas Time：Meas Time：sampled at interval equal to Meas time.
  - 100ms ：sampled at interval 100ms (0.1s).
  - 200ms ：sampled at interval 200ms (0.2s).
  - 1s ：sampled at interval 1s

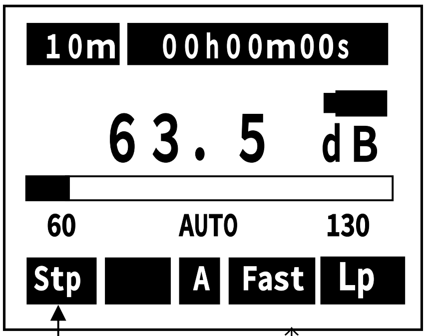
  ** Meas Time is time set with [Meas Time] key (1s～-•••).

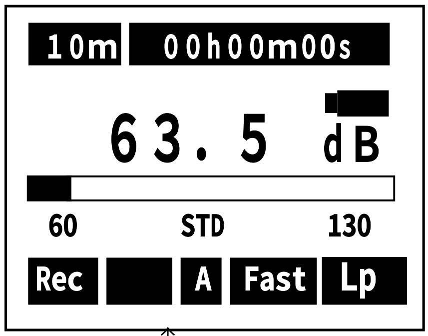
  Fixed to Mease Time, when when RSR card is installed,

  ＊ Select 10s or more in L_{Atm5} measurement.
- Sta ： Registers the starting time for recording (YY/MM/DD HH/MM/SS) (Year/Month/Date, date time/minute/second).
- Stp ： Registers the stop time for recording (YY/MM/DD HH/MM/SS) (Year/Month/Date, date time/minute/second)

NOTE
Measurement starts when the selected level is exceeded after the time specified with [Sta Time], In the following example : ：When the level exceeds 65dB after 18:16 October 10, Recording starts and the measurement is made once during the time specified with [Meas Time]. Recording is continued, in Interval Repeat mode, until the level falls or until 20:16 October 12,. Data is shown according to System Mode.

< Standby screen >

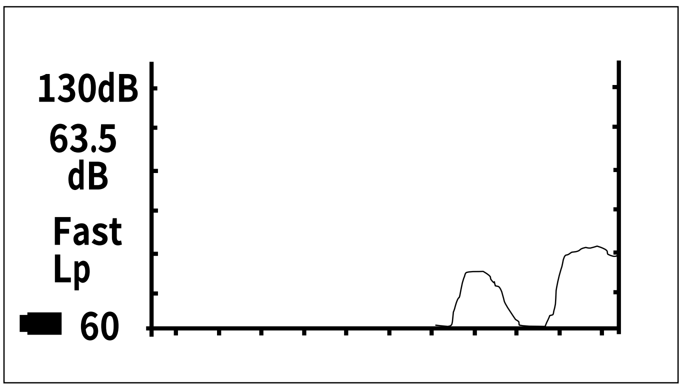




Default screen   Screen with STD card installed

The Stp blinks when [Start Stop] Key is pressed to confirm the stand-by state.

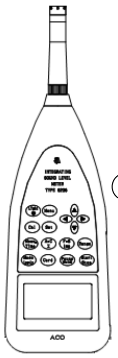
< Display >



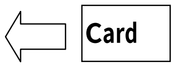
How to use the Memory Card (SD Card - Standard -)

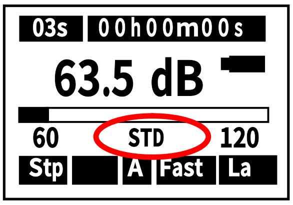
The measured data can be stored on the memory card (SD card) to be processed by personal computer. When the memory card (SD card) is inserted, display【STD】blinks.

Card Installation

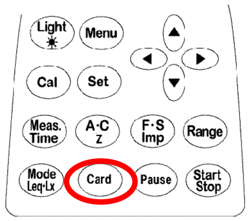


① Insert the a card.

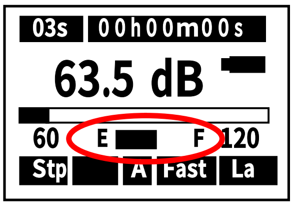




②STD blinks.

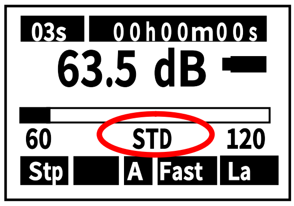


③Press [Card] key.



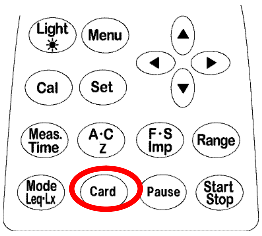
④The remaining battery capacity displayed.

After 3s

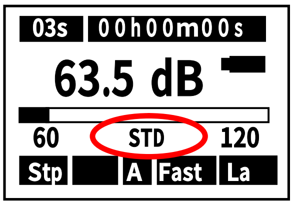


⑤Lighted STD is a standby display on starting the measurement.

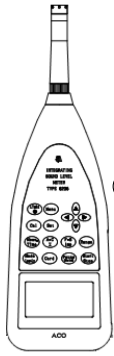
Eject the card



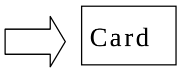
①Press Card key.



②STD blinks.



④Eject the Card.



Measurement

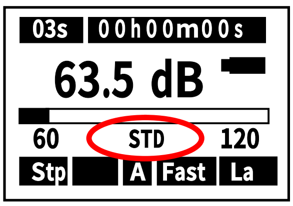
Ref. Measurement Procedure

Press Start key

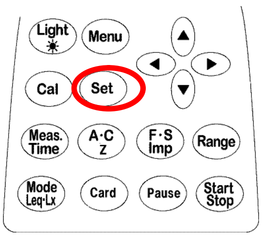
<E.g. Acquired data to be stored in CSV format.

Measur day	time	weight	Range	Time set	LAeq	LAE	Lmin	
2009/04/13	18:17:10	F	80dB	000h00m03s	49.8	54.6	40	…
2009/04/13	18:17:13	F	80dB	000h00m03s	56.6	61.3	47.4	…
2009/04/13	18:17:16	F	80dB	000h00m03s	66	70.7	51.9	…
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Delete the card data



① Confirm STD blinking.



②Keep [Set] Key pushed for a few seconds in the situation with the card installed.

Data all Clear
Yes＝Start
No＝ Set

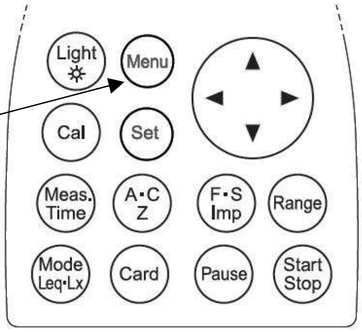
Sure ?
Yes＝Start
No＝ Set

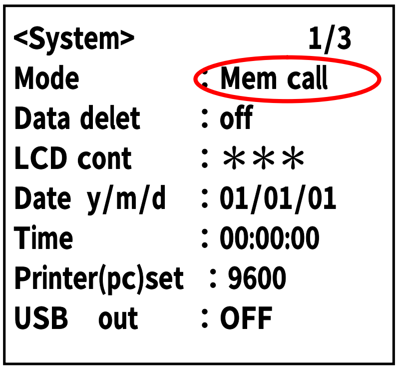
③ Delete all the data along the displayed flow of operation, then to return to the former window.

Data recall from the memory

＜ Operation＞

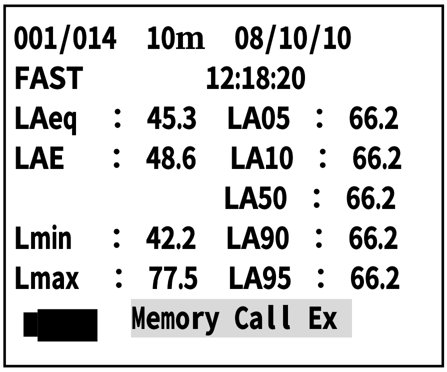
Menu key





The [Mode: Normal] in <system>1/3 of [Menu] screen is changed to [Mode: Mem call] with ▲▼key, which then leads to the memory display screen by pressing [Set] key. .

<Display>



← Memory No., Sampling time, Calendar
← Time constant,　Date/time of the Measurement
← Symbol *** refers to the item which is set OFF in Menu screen.　.

← Display blinks

Select the data with ▲▼ key, accelerating by keeping on pushing the cursor
On pressing Card key in the state of Memory Call, it changes to Memory Call Ex and the data in the Card is displayed

Memory Call IN：Data stored in the internal SRAM.

Memory Call Ex：Date stored in the external SD_Card

[Start/Stop] key pressed starts printing the displayed data (data communication).
※To return to the default screen, change [Mode: Memory call] in < System>1/3 of Menu screen to [Mode :Normal] with ▲▼ key.

Print/Data management

Print

This equipment is provided with a function of printing the measured data with a specified serial printer.

```
<System>              1/3
Mode            : Print
Data delet      : off
LCD cont        : ＊＊＊
Date  y/m/d     : 01/01/01
Time h/m/s      : 00:00:00
Printer(pc)set  : 9600
USB    out      : OFF
```

← Set to
Mode: Print

＜Parameter setting＞

1) Turn off POWER switch and connect the printer before turning on **POWER** switch again.
2) Select Print in [Menu] < System > 1/3 screen and press [Set] key.
3) Then, the following screen appears.

＜Display＞

```
001/014   10m   08/10/10
FAST               12:18:20
LAeq  :  45.3    LA05  : ***.*
LAE   :  48.6    LA10  :  66.2
                 LA50  : ***.*
Lmin  :  42.2    LA90  : ***.*
Lmax  :  70.5    LA95  : ***.*
          Print   IN
```

← The top data

← Display blinks

elect the data with ▲▼key, accelerating by keeping on pushing the cursor.
On pressing [Card] key in the state of Print, it changes to Print Ex and the data in the Card is displayed

Memory Call IN：Data stored in the internal SRAM.
Memory Call Ex：Date stored in the external SD_Card

Print (date communication) starts with the top of the data by pushing the Start/Stop key.
Printing can be paused with [Pause] key and restarted by pressing it again from the current data line. .
The printing is canceled by pressing [Stop] key and stands by at the top data display screen.
By pressing [Menu] key, the screen moves to Menu display.
By changing [Mode;Print] to [Mode;Nomal] with ▲▼key and pressing [Set] key, the screen moves to the memory display.

Saving Data to PC

This equipment is provided with data saving function using the specified data management software.

Data management with USB port

<System>	1/3	
Mode	: PC out	← Set Mode to PC out display
Data delet	: off	
LCD cont	: ＊＊＊	
Date y/m/d	: 01/01/01	
Time h/m/s	: 00:00:00	
Printer(pc)set	: 9600	
USB out	: OFF	

＜Operation＞

Select PC out in [Menu] < System > 1/3 screen and press [Set] key.

＜Display＞

001/014 10m	08/10/10			← The top data
FAST	12:18:20			
LAeq :	45.3	LA05	: ***.*	
LAE :	48.6	LA10	: 66.2	
		LA50	: ***.*	
Lmin :	42.2	LA90	: ***.*	
Lmax :	70.5	LA95	: ***.*	
	PC out			← Display blinks.

Select the data with ▲▼key, accelerating by keeping on pushing the cursor
On pressing Card key in the state of Print, it changes to Print Ex and the data in the Card is displayed

Memory Call IN：Data stored in the internal SRAM.
Memory Call Ex：Date stored in the external SD_Card

Date communication starts with the top of the data by pushing [Start/Stop] key.
Printing can be paused with [Pause] key and restarted by pressing it again from the current data line. .
The printing is canceled by pressing [Stop] key and stands by at the top data display screen.

Example of file creation

The file is created as follows. :

When A characteristics (Time constant F and S) is selected:

・Single

001.csv ← ：Whenever [Start] key is pushed, this single data line is made. (a single data since the mode is Single.)

002.csv ↓

・

・

Meas…day/Meas…time/Time weight/Level Range/Time sett/LAeq/Lmin……./LA95							
2009/03/2	9:54:52	F	80dB	000h…10s	48.9	42.3	43.9

・Repeat

001.csv ← ：Whenever [Start] key is pushed, this data line is made. (two or more data/file)

002.csv ↓

・

・

Meas…day /Meas…time/Time weight/Level Range/Time sett/LAeq/Lmin……./LA95							
2009/03/2	9:54:52	F	80dB	000h…10s	48.9	42.3	43.9
2009/03/2	9:54:52	F	80dB	000h…10s	48.9	42.3	43.9
2009/03/2	9:54:52	F	80dB	000h…10s	48.9	42.3	43.9
・							
・＜as many data lines as indicated by Repaet＞							
2009/03/2	9:54:52	F	80dB	000h…10s	48.9	42.3	43.9

Eventually, in the card, Single and Repeat data files are created at random.

＜Example＞

001.csv ←：File made in single mode (1 data/ 1file)
002.csv ←：File made in single mode (1 data/ 1file)
003.csv ←：File made in repeat mode (two or more data/file)
004.csv ←：File made in single mode (1 data/ 1file)
005.csv ←：File made in repeat mode (two or more data/file)

・

・

↑ At most 999 CSV files can be made, where [Start] key is pressed 999 times..

Only L_{AI}eq is made.

Meas…day /Meas…time/Time weight/Level Range/Time sett/LAIeq					
2009/03/2	9:54:52	I	80dB	000h…10s	52.3
2009/03/2	9:54:52	I	80dB	000h…10s	52.3
2009/03/2	9:54:52	I	80dB	000h…10s	52.3
・.					
＜as many data lines as indicated by Repaet＞					
2009/03/2	9:54:52	I	80dB	000h…10s	52.3

Output terminal

AC、DC Output

AC Output

The AC Output is the frequency-weighted signal.
Output: 1Vrms (FS), Output impedance: 600Ω, Load impedance > 10kΩ

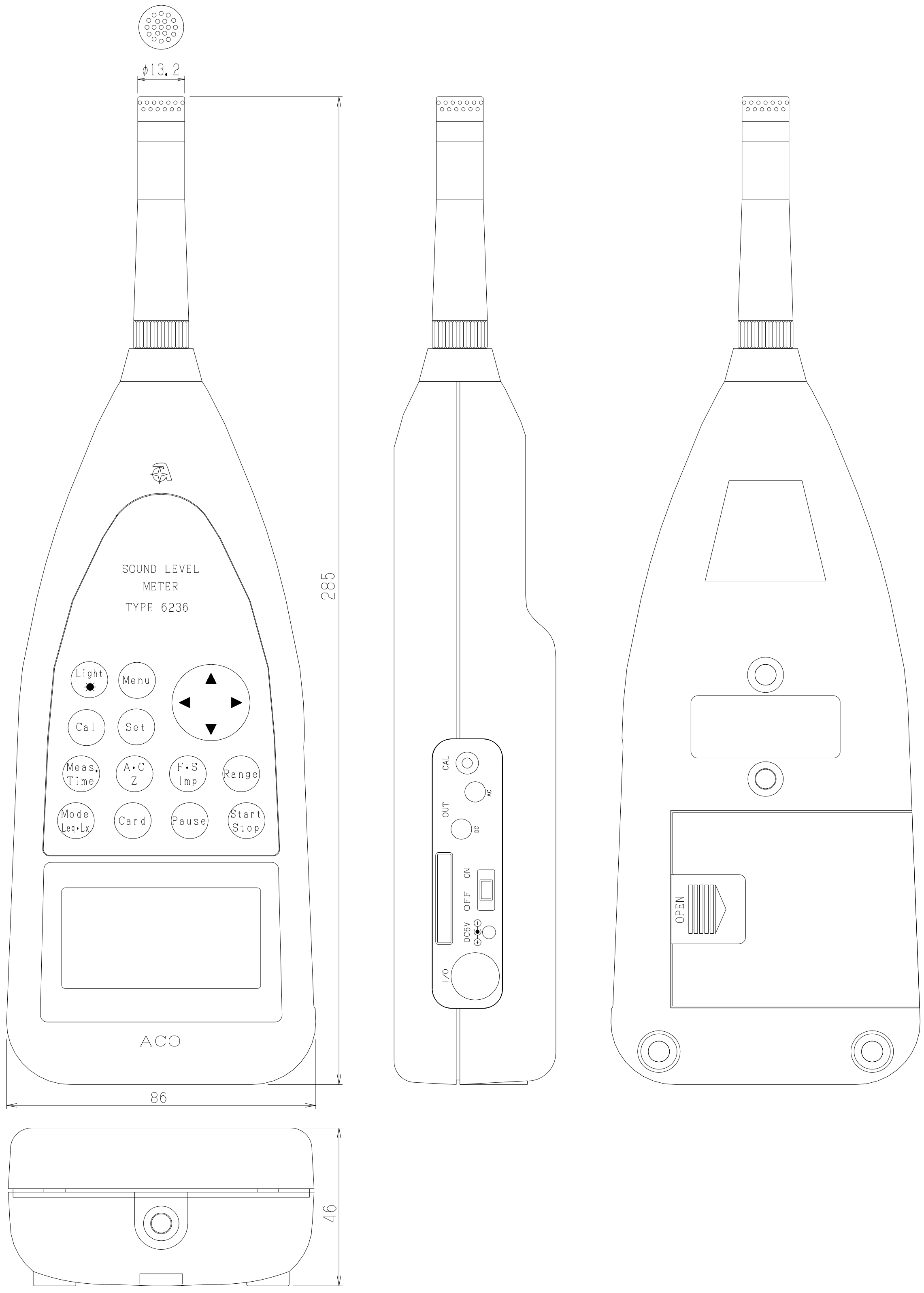
DC Output

The DC Output is the frequency-weighted, root-mean-square-detected, and then logarithmic converted signal.
Output: 2.5V (FS), 0.25V/10dB, Output impedance: 50Ω, Load impedance > 10kΩ

Specification

1) Type	：TYPE 6236
2) Description	：Sound Level Meter
3) Applicable Standards	：JIS C1509-1：2005 ClassⅡ, IEC 61672-1：2002 Class Ⅱ
4) Frequency Range	：20Hz～20kHz (Conforms with measurement law 20Hz～8kHz)
5) Microphone(Sensitivity)	：TYPE 7052NR(-33dB, Stand-alone-31dB)
6) Level Range Control	：10dB 6step 20～80dB、20～90dB、20～100dB、20～110dB、30～120dB 40～130dB
7) Measurement Level	A ：28～130dB (0～80dB／0-dB measurement function in ON) C ：36～130dB Z(FLAT) ：40～130dB C peak ：55～141dB Z（FLAT） peak：60～141dB
8) Self-noise level	：The lower limit of the measurement range in dB lies 6dB higher than self-noise level.
9) Linearity Range	：100dB
10) Time weighting	：Fast、Slow、Impulse
11) Frequency weighting	：A, C, Z（FLAT）
12) Measurement items	：Sound pressure level(Lp), A-weighted sound pressure level, C-weighted sound pressure level(L_A, L_C),Equivalent continuous A-weighted sound pressure level(L_{Aeq}), Sound exposure level(L_{AE}), Maximum sound pressure level(L_{Amax}), Minimum sound pressure level(L_{Amin}), Percentile sound pressure level(5 freely selectable values, L_{AN}), Peak sound pressure level(L_{peak}), C-weighted peak sound pressure level (L_{Cpeak}), C-weighted equivalent continuous sound pressure level(L_{Ceq}) Power average of maximum sound pressure level in a given interval(L_{Atm5}), Impulse sound pressure level(L_{AI}), Impulse equivalent continuous sound pressure level(L_{AIeq})
13)Measurement time	：1s, 3s, 5s, 10s, 1mim, 5mim, 10mim, 15mim, 30mim, 1h, 8h, 12h, 24h, Manual(Max. 199h59m59s)
14) Sampling Time	：20.8μs（L_{eq}、L_{max}、L_{min}), 100ms(L_N)
15) Data clear function	：Pause, and a function that deletes preceding 3 or 5 sec. data. Memory start ；Selectable Auto or Manual
16) Timer function	：A marker can be set to start and stop the measurement at any specified moments.
17) Display	：Liquid crystal and Backlight(128×64 points)
Display range	：4digits display
Display cycle	：display Period: 1s
Bar display	：display Period: 0.1s
Warning	：Over ；＋3dB from upper limited scale Under；－0.6dB from lower limited scale
Battery display	：5 steps display
Date	：year/month/day/ hour：minute：second

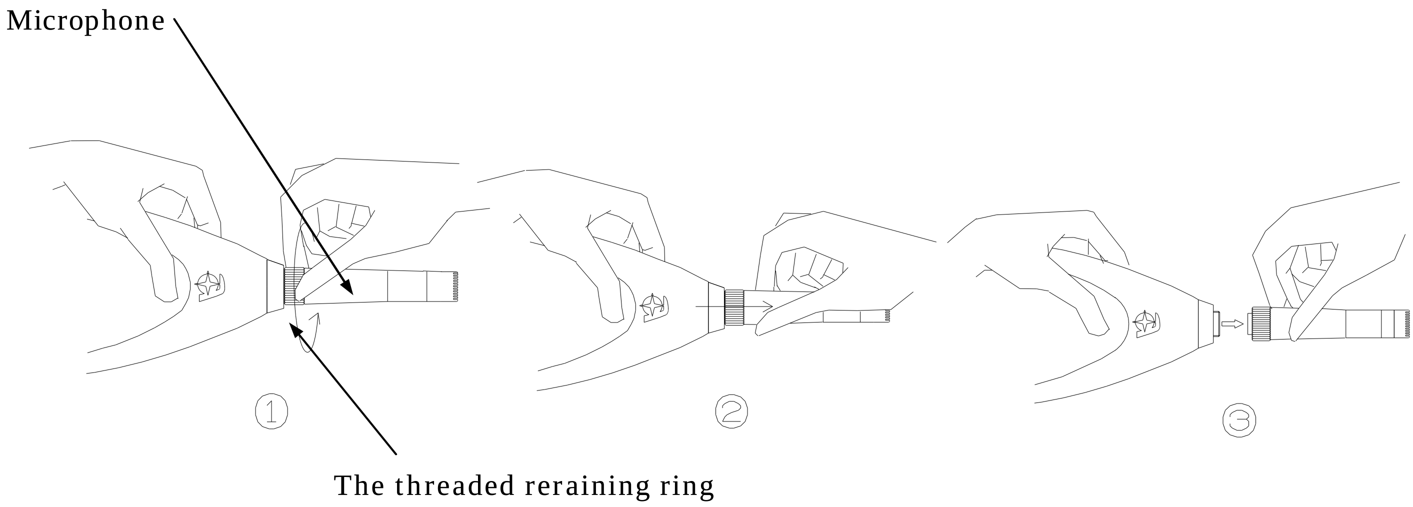
18) Calibration signal	: Electric calibration with internal oscillator (1kHz sine wave)
19) AC output	: φ2.5 Jack
Output	: 1Vrms (FS)
Output impedance	: 600Ω
Load impedance	: more than 10kΩ
DC output	: φ2.5 Jack
Output	: 2.5V (FS), 0.25V/10dB,
Output impedance	: 50Ω
Load impedance	: more than 10kΩ
20) RMS detection circuit	: True RMS detection circuit (computing type)
21) Processing	: Digital
22) Pause	: Normal pause function, as well as the function of canceling the data before pausing the measurement, are available.
23) Data Storage Functions	: Sound pressure level or Processed values stored in built-in Memory or Memory card
Manual Storage	: Sound level, Calculation value, Memory time, Store the Sampling Time to Built-in memory or on Memory card
Auto Storage	: Sampling interval 100ms, 200ms, sound level , Leq etc.
Processing Card	: Storage of calculation results
24) I／O	: Direct output to printer, control and output data to computer Digital output of real-time noise waveform with USB interface
25) Comparator Output	: Comparator Function with threshold level
26) Battery Type	: Four 1.5V Alkaline cells IEC type LR6, Optional AC adapter
Battery life	: Alkaline dry cell ; Approx.20hours when Switch on a back light ; Approx.1/3
Consumption current	: Approx.140mA (When input 6V) at Calculaiotn OFF.
27) Operating temperature	: −10～50℃ 30%～90%RH（no condensation）
28) Size	: 86 (W) ×285 (H) ×46 (D)
29) Weight	: Less than Approx.450g (Including batteries）



Appearance diagram of Sound Level Meter TYPE 6236

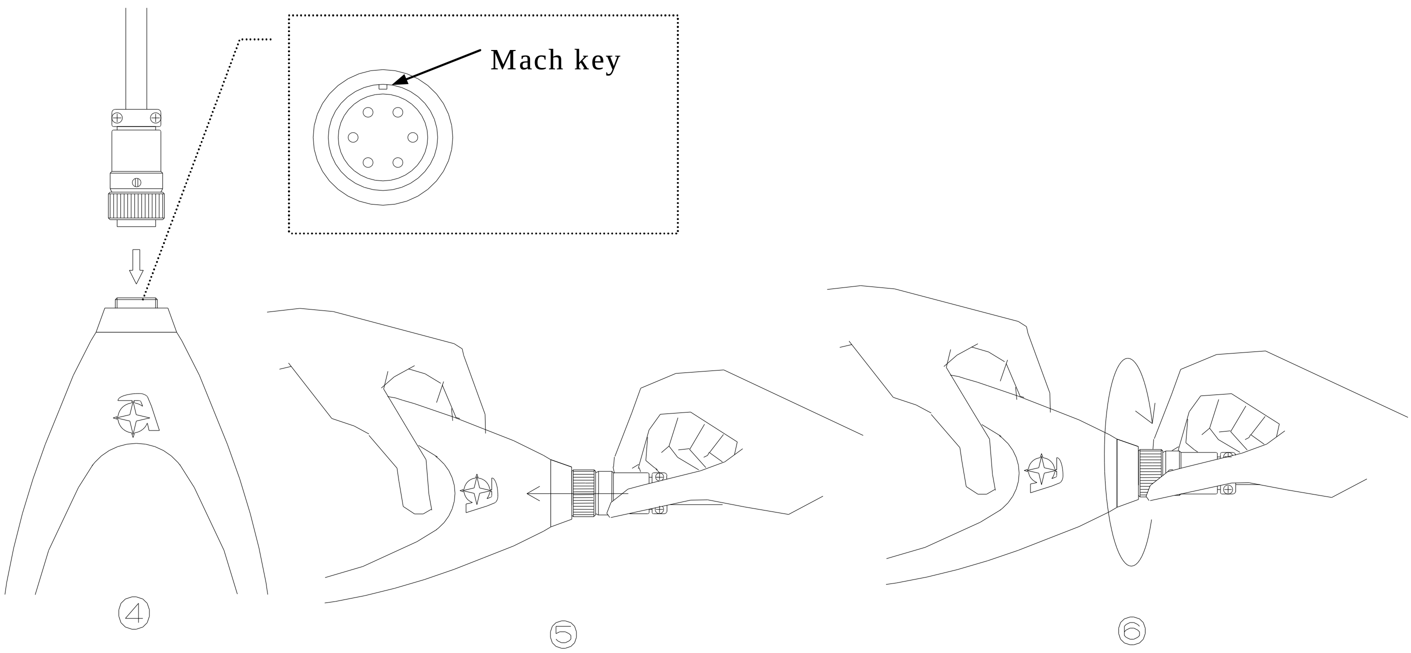
Pin Connections and How to Connect the Extension cable

1) Detach microphone from the body of the meter.



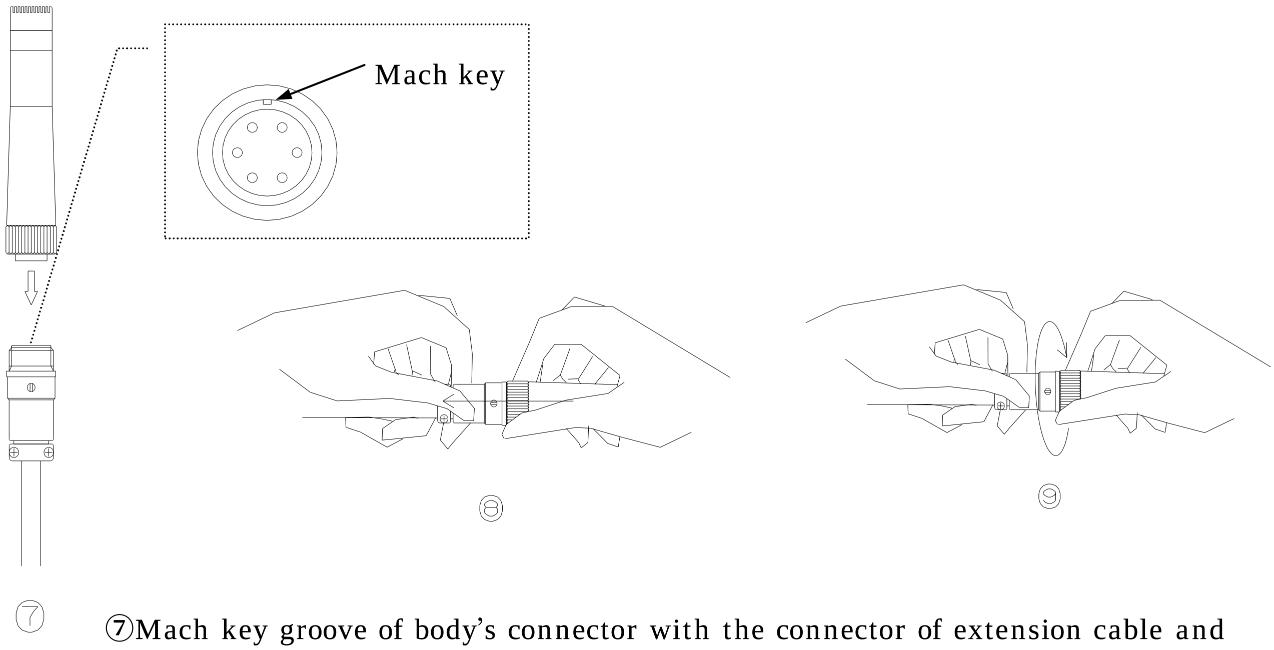
① Turn the threaded reraining ring a little to the left.
② Pull out microphone as shown.
③ Repeat ① turn a left and ② pull out a little 5-8 times and you can separate.

2) Then plug the male connector of extension cable into the connector of the body.



④Mach key groove of body’s connector with the connector of extension cable and insert.
⑤Push the connector of extension cable.
⑥Turn the threaded reraining ring a little as shown repeat ⑤ and ⑥ 5-8 times and you can connect.

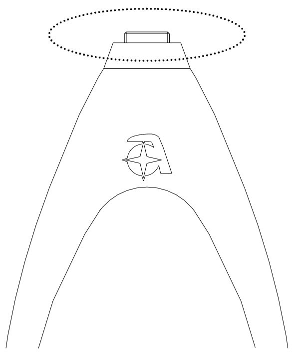
3) Attach microphone to the female connector of extension cable.

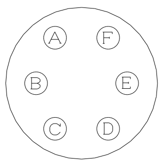


⑦Mach key groove of body’s connector with the connector of extension cable and insert.
⑧Push the connector of extension cable.
⑨Turn the threaded reraining ring a little as shown repeat ⑧ and ⑨ 5-8 times and you can connect.

※ **Note; Do not turn only the threaded reraining ring connecting.**
It causes damage to the connector.

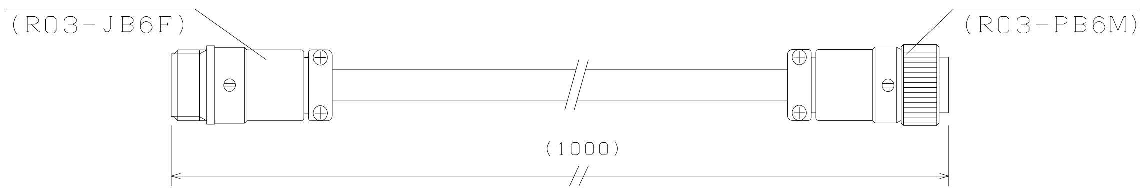
【Wiring diagram of Main body side connector】

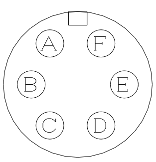




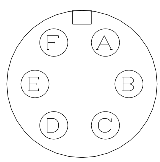
A；＋6.5V　E；NC
B；NC　F；SIG
C；－6.5V
D；GND

【Wiring diagram of Extension cable】





A；＋V　D；COM
B；NC　E；NC
C；－V　F；SIG



A；＋V　E；NC
B；NC　F；SIG
C；－V　Case ; Shield
D；COM